

The Intergenerational Transmission of Employers and the Earnings of Young Workers*

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Abstract

This paper investigates how parental connections to firms shape early-career earnings. I use employer-employee linked data to study one important type of connection: jobs obtained at a parent's employer. Twenty-nine percent of individuals work for a parent's employer at least once by age 30. Exploiting transitory, firm-specific fluctuations in hiring conditions at the parent's employer, I estimate that working for a parent's employer increases initial earnings by 24 percent, primarily by providing access to higher-paying firms. Individuals with higher-earning parents are more likely to work for a parent's employer and experience larger earnings gains when they do. Consequently, the elasticity of initial earnings with respect to parental earnings would be 12 percent lower if no one found a job through these connections. The findings suggest that connections to firms through one's social network may be an important determinant of intergenerational mobility.

Keywords: intergenerational mobility, labor market networks, job ladders

JEL Codes: J31, J62, L14

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1 Introduction

To what extent do parental connections in the labor market shape the earnings of young workers? A large literature studies how parents affect their children’s long-run outcomes (Black and Devereux, 2010), focusing primarily on childhood environments such as neighborhoods (Chetty et al., 2026a) and parental income (Bulman et al., 2021). Yet parents may also directly influence their children’s earnings in adulthood by helping them find jobs through their labor-market networks. This channel may be important because most jobs are found through personal connections (Ioannides and Datcher Loury, 2004), and the firms at which workers are employed are a major determinant of their earnings (Card et al., 2018).

This paper studies how parental connections shape the earnings of young workers by focusing on one specific channel: jobs obtained at a parent’s employer. Working for the same firm as a parent—i.e., the *intergenerational transmission of employers*—is not uncommon: I find that 29 percent of individuals work for the same firm as a parent by age 30, which is consistent with prior evidence (Kramarz and Skans, 2014; Corak and Piraino, 2011; Stinson and Wignall, 2018). I use linked employer-employee data from the Longitudinal Employer-Household Dynamics (LEHD) program and the 2000 Decennial Census to measure how often young workers obtain jobs through these connections and estimate the causal effect on earnings. I then examine how the benefits vary with parental income and quantify the implications for the intergenerational persistence in earnings. The estimates of the magnitude and source of the earnings gains shed light on how and why connections, more broadly defined, may shape income inequality.

I begin by showing that 5 percent of individuals work for a parent’s employer in their first job, with substantial differences by parental income. Among children of parents in the top percentile of the earnings distribution, 7 percent work for a parent’s employer, compared to just 2 percent among those from the bottom percentile. The analysis focuses on the first stable job, which plays a central role in shaping long-term career trajectories (Kahn, 2010; Arellano-Bover, 2024).¹ However, these patterns are not limited to the start of the career: individuals continue to work with their parents in

¹The definition of the first job is provided in Section 2.

subsequent jobs, and disparities by parental income widen over time.

Parental connections are one explanation for why children and parents may work for the same firm. However, parents and children may also sort into the same firms due to other reasons, such as shared skills or preferences. To isolate the role of connections, I compare the probability that a child works for the firm their parent recently joined and currently works for to the probability of working for a firm the parent will join in the near future. Children are five times more likely to work for a parent's current employer than for a parent's future employer. Assuming the only systematic difference between the two firms is the presence of a connection, this implies that approximately 80 percent of individuals who work for a parent's employer do so as a result of parental connections. This may understate the role of connections if parents also have ties to future employers—for example, through extended family or friends.

The main empirical challenge is to estimate the earnings consequences: among individuals who work for their parent's employer, how much more do they earn relative to their outside option? Estimating this causal parameter is difficult because those who work for their parent's employer may differ in unobserved ways. Indeed, even when comparing siblings, I find that those without a college degree are twice as likely to work for their parent's employer as their college-educated siblings. An empirical strategy must account for the fact that young workers with limited outside options are more likely to rely on parental connections.

The ideal experiment to estimate the causal effect of working for a parent's employer would randomly prohibit some firms from hiring the children of current employees. Random assignment across firms would serve as an instrument, generating variation in access to jobs at a parent's employer that is orthogonal to other determinants of child outcomes. This design would identify the average effect for individuals who work for their parent's employer and would allow for the possibility that some children would choose not to work for their parent's firm, and some firms would choose not to offer them a job.

I approximate the ideal experiment using an instrumental variables strategy that exploits transitory, firm-specific fluctuations in hiring at the parent's employer. The design compares children whose

parents work at the same firm but who enter the labor market at different times. Children entering the labor market when their parent's firm happens to be hiring fewer workers are less likely to begin their careers at that firm. To isolate variation that reflects firm-specific shocks rather than broader labor market trends, I net out hiring fluctuations among other firms in the same commuting zone and industry. Thus, identification comes from changes over time in hiring at a parent's employer relative to other firms in the same local labor market.

I implement this design by instrumenting for whether an individual finds their first job at a parent's employer using the hiring rate at that firm. The main specification controls for fixed effects for the parent's employer, fixed effects for the local labor market (defined by commuting zone, industry, and quarter), and parental earnings. To illustrate that the approach isolates transitory, firm-specific fluctuations in hiring, I show that child outcomes are strongly related to contemporaneous hiring rates at the parent's employer, but unrelated to hiring rates just a few years earlier or at other observably similar firms.

I find that working for a parent's employer leads to a 24 percent increase in initial earnings at a young worker's first job. These gains are substantial—approximately one-third the size of the college wage premium—and are generally larger for those from high-income families. The effects are also persistent: individuals who began their careers at their parent's employer earn 17 percent more three years later.

These gains are largely explained by parents providing access to higher-paying firms in the blue-collar sector for children who would otherwise have worked in the unskilled service sector. I estimate firm-specific pay premiums using the AKM decomposition of earnings (Abowd et al., 1999) and find that the impact on the AKM firm effect is virtually identical to the impact on individual earnings. In other words, parents provide access to firms that pay all workers higher wages, not to relatively high-paying jobs within these firms.

Lastly, I show that individuals with higher-earning parents benefit more from parental connections because they are more likely to work for a parent's employer and experience larger earnings gains when they do. Combining the descriptive and causal estimates, I quantify the implications of these

connections for intergenerational mobility. The elasticity of children's initial earnings with respect to parental earnings would be 12 percent lower in a counterfactual world in which no one worked for a parent's employer. These results suggest that part of the intergenerational persistence in earnings arises because parents use workplace connections to help their children access higher-paying firms.

This paper builds on and relates to several literatures. First, this paper contributes to the literature on intergenerational mobility, which seeks to explain why economic outcomes persist across generations (Black and Devereux, 2010). Prior work shows that children from low-income families experience greater upward mobility in areas with more cross-class social connections (Chetty et al., 2022) and higher parental employment rates (Chetty et al., 2026b). However, the underlying mechanisms remain unclear: social connections may shape children's development and aspirations, or they may directly improve access to jobs through referrals. Related work shows that parental connections contribute to earnings gaps between ethnic groups (San, 2022), and that family background is associated with children's workplaces (Dobbin and Zohar, 2023), occupational choices (Haeck and Laliberté, 2025), and post-displacement earnings (Krolikowski et al., 2020; Kaila et al., 2024). This paper shows that one specific connection—access to a parent's employer—shapes early-career outcomes and intergenerational mobility, suggesting that access to firms through social networks, more broadly defined, may be an important driver of economic mobility.

Second, this paper relates to the literature on labor market networks. Prior research finds that social connections shape labor market outcomes, but there is little consensus on the magnitude or source of the earnings gains from finding a job through a social contact (Bayer et al., 2008; Hellerstein et al., 2011; Schmutte, 2015; Dustmann et al., 2016; Zimmerman, 2019; Caldwell and Harmon, 2019; Barwick et al., 2023). I build on descriptive evidence documenting the tendency of children and parents to work at the same firms (Kramarz and Skans, 2014; Corak and Piraino, 2011; Stinson and Wignall, 2018) and estimate the causal effect of obtaining a job through these connections. The results show how the use of connections can lead to large earnings gains by facilitating access to higher-paying firms.

Third, this paper contributes to the literature on how firm-level pay policies shape earnings

inequality. The AKM literature documents large differences in pay across employers, but there is an ongoing debate about whether these gaps reflect firm-level pay premiums or arise from the endogenous sorting of workers across firms (Card et al., 2018). Prior research addresses this challenge by studying displaced workers who separate from their employers for reasons plausibly unrelated to their earnings potential, finding that earnings changes closely track changes in firm pay premiums (Lachowska et al., 2020; Schmieder et al., 2023). I provide complementary evidence on the importance of firm pay premiums by exploiting quasi-experimental variation in the firms that young workers join.

The paper is organized as follows. Section 2 describes the data. Section 3 examines how often and why individuals work for their parent’s employer. Section 4 estimates the earnings effects. Section 5 quantifies implications for intergenerational mobility, and Section 6 concludes.

2 Data

I rely on two main sources of data: (1) the 2000 Decennial Census and (2) the LEHD program. The Decennial Census is a household survey that allows me to measure relationships between parents and children living in the same household in 2000. In principle, these data cover all individuals living in the United States. In practice, some individuals are not surveyed, and nonrespondents are more likely to be members of minority groups and low-income households.

The LEHD is a linked employer-employee dataset produced by the U.S. Census Bureau and measures the earnings and employers of both parents and their children between 1990 and 2018. The LEHD is constructed from two core administrative datasets: (1) unemployment insurance (UI) records, which provide job-level earnings records; and (2) the Quarterly Census of Employment and Wages, which provides establishment-level characteristics. These data capture roughly 96 percent of private non-farm wage and salary employment in the United States but do not cover self-employment. Coverage begins at different times across states. By 2000, the data cover 42 states; by 2005, they cover all states except Massachusetts (see Figure A.1).

My sample frame includes individuals for whom I can measure parent-child relationships and

early-career outcomes. The sample frame includes children in the 2000 Decennial Census who (1) lived with a parent, (2) are expected to graduate high school between 2000 and 2013, and (3) reside in a state that began reporting to the LEHD at least two years prior to the expected year of high school graduation.² 91 percent of individuals younger than 18 live with a parent in 2000. By the end of 2018, the youngest and oldest individuals in the sample were 23 and 37 years old, respectively. There are approximately 48 million individuals in the sample frame.

I drop individuals from the sample if I am unable to link them across datasets or accurately measure parental earnings. Individuals are identified by a Protected Identification Key (PIK), which the Census Bureau generates using personally identifiable information. I drop 19 percent of the sample frame because the child is not assigned a PIK and therefore cannot be linked to the LEHD. I drop an additional 7 percent because a parent is not assigned a PIK or the household in the Decennial Census contains more than 15 individuals. Some individuals with very low earnings have earnings from other sources not covered by the LEHD. Thus, I drop an additional 7 percent of the sample frame if the combined annual earnings of the parents are less than \$15,000 (I discuss the measurement of parental earnings in more detail below). Of the 48 million individuals in the sample frame, approximately 32 million, or 67 percent of the sample frame, meet these restrictions (see Table A.1). The resulting sample is broadly representative of families for whom wages constitute the majority of earnings and income, a group that excludes the very poor (approximately the bottom 10 percent of households) and the very rich (approximately the top 1 percent of households).

First stable job. I define the first stable job, or simply the first job, as the first quarter in which an individual earns at least \$3,300 per quarter—which corresponds to working approximately 35 hours per week at the federal minimum wage—in the current and the two subsequent quarters, and receives positive earnings from the same employer for those three quarters.³ Conceptually, this is the first quarter in which work becomes the primary activity. I refer to this employment

²Expected year of high school graduation is based on month and year of birth and individuals born between September 1st and August 31st are assigned to the same cohort. The sample frame includes individuals born between September 1st of 1981 and August 31st of 1995. When measuring parent-child relationships, I include biological, adopted, and step children.

³Dollar values are converted to 2016 dollars using the Consumer Price Index for All Urban Consumers.

spell as the *first stable job* and measure *initial earnings* during the first full quarter of employment at this job.⁴ Approximately 26 million individuals, or 82 percent of those who meet the sample restrictions, obtain a first stable job by the end of 2018. Individuals who never find a first stable job have persistently low earnings, with average annual earnings of only \$1,130 at age 30.

Three pieces of evidence suggest that my definition of a first stable job is reasonable. First, individuals experience a dramatic and persistent increase in earnings when at their first job. Average annual earnings increase from \$7,084 to \$29,080 when the first job begins (see Figure A.2). Second, the age at first job agrees with other common definitions of labor market entry. In my data, 86 percent of young workers find their first job between ages 18 and 26. I calculate an analogous measure using the NLSY97 and find that 86 percent of respondents find their first stable job between these ages.⁵ Furthermore, 83 percent of workers in the NLSY97 data are not enrolled in school when they find their first job, which suggests that my measure is not primarily picking up jobs held by students. Third, 40 percent of young workers remain at their first employer for at least three years.

Parental earnings. Without data on the full labor market history, a common approach is to calculate parental earnings as the average earnings over a limited number of years. However, the LEHD presents unique challenges as there is no way to distinguish between zero earnings and earnings that are not covered by the LEHD frame. To account for these issues, I construct a long-run measure of parental earnings by regressing quarterly earnings on an individual fixed effect and a third-degree polynomial in age within samples defined by the interaction between state of residence in 2000, sex, and education. Using these parent-specific age-earnings profiles, I calculate the average annual earnings between the ages of 35 and 55. Parental earnings are the sum of the individual earnings of both parents. I calculate percentile ranks based on parental earnings within cohorts defined by expected year of high school graduation. See Appendix B.1 for details.

Employers. Employers are identified by the state-level employer identification number (SEIN),

⁴A full-quarter employment spell occurs when a worker receives strictly positive earnings from the same employer in the current, previous, and subsequent quarter.

⁵Figure A.3 presents the distribution of age at first job for individuals in my sample and in the NLSY97. The analogous measure constructed from the NLSY97 is the first time an individual works at least 35 hours for 36 consecutive weeks (three quarters). An alternative approach is to focus on labor market outcomes after all schooling is completed, and Figure A.3 also presents results for this measure.

which typically captures the activity of a firm within a state and industry.⁶ I use “firm” and “employer” to refer to the entity identified by the SEIN. About half of individuals work for a firm with multiple establishments, and the LEHD imputes the link between workers and establishments. I primarily focus on the firm, but in some analyses I use the imputed establishment to measure the location of the job within a state.

3 Use of Parental Connections

I begin by documenting how common it is to work for a parent’s employer. Columns 1 through 3 of Table 1 present summary statistics for the full sample and for individuals who do not and do work for a parent’s employer, respectively. Five percent of individuals work for a parent’s employer at their first job, and these individuals tend to stay at these jobs longer, are less likely to be employed in the unskilled service sector and more likely to be employed in the blue-collar sector, and earn slightly less.⁷ Of individuals who work for a parent’s employer, only 19 percent have a parent who is in the top percentile of the within-firm earnings distribution, suggesting that the majority of these parents are regular employees, not owners or top executives.

Figure A.5 plots the cumulative share of individuals who work for a parent’s employer between the ages of 16 and 30. By age 30, 29 percent of individuals have worked for a parent’s employer. This is consistent with Stinson and Wignall (2018), who find that 22 percent of sons work with their fathers by the time they are 30 years old.

Parental connections offer one explanation for why someone may work for the same firm as their parent. Connections could facilitate access to these jobs by reducing information frictions between workers and firms or influencing hiring decisions through referrals or favoritism. However, some individuals might work for the same firm as a parent for other reasons. For example, parents and children might have similar skills and live in the same area, which could lead them to sort into the same firms. To distinguish between these two explanations, we need to know how often individuals

⁶A worker could have positive earnings at multiple employers in a given quarter. In such cases, I measure the characteristics of the employer providing the most earnings in that quarter.

⁷I group two-digit North American Industry Classification System (NAICS) industry codes into three sectors: unskilled services, skilled services, and blue-collar. See Appendix B.2 for details.

would work for their parents' employers absent connections.

I answer this question by comparing the likelihood of working for a parent's current and future employer, since these firms are likely similar along many dimensions except that parents are less likely to have connections at the future employer. I identify parents who begin a new job within three years of their child entering the labor market. Figure 1 plots the proportion of children who work for that firm against the quarter in which their parent started the job. The sample is restricted to parents who remain at the firm for at least three years, ensuring that if they joined the firm before their child entered the labor market, they are still at the firm when the child finds their first job. Individuals are five times more likely to work for a firm that their parents joined two to three years before the child finds their first job compared to two to three years after.

If parental connections are the only systematic difference between current and future employers, these estimates suggest that 80 percent ($0.8 = 1 - \frac{1}{5}$) of individuals who work for a parent's employer found their job via parental connections. This likely overstates the likelihood that an individual finds a job at their parent's employer for reasons unrelated to connections, since parents may have ties to the future employers through other social contacts such as extended family or friends.

I also compare the likelihood of working for a parent's employer with the likelihood of working for an observably similar firm in the same local labor market. For each employed parent, I identify ten other firms in the same commuting zone, two-digit industry, and size class (greater or less than 500 employees). Six percent of individuals work for their parent's employer at their first job, while only 0.03 percent work for the other firms, on average.⁸ In other words, individuals are 200 times more likely to work for a parent's employer than for a similar firm in the same local labor market. Furthermore, Table 1 shows that 70 percent of individuals who work for a parent's employer live in urban areas and they are no more likely to work for large firms, suggesting that the patterns are not driven by cases in which a single employer dominates a local labor market.

Figure A.6 shows that working for a parent's employer is more common in industries where labor market networks are used more often. Using responses to the first wave of the NLSY97, I calculate

⁸Across the 10 draws of firms, the minimum and maximum percentages of individuals who work for one of these other firms are 0.033 and 0.035 percent, respectively.

the proportion of individuals who were hired by or recommended for their job by a parent and plot this statistic against the proportion who work for a parent's employer by industry. The correlation is positive, with a regression coefficient of 2.9 (p-value=0.003). Both measures suggest that social connections are less commonly used in the unskilled service sector than in the blue-collar sector.

Taken together, these results suggest that individuals who work for a parent's employer do so primarily because of parental connections. This is consistent with Loury (2006), who finds that 10 percent of young men found their current job through a parent, as well as other research emphasizing the role of social connections in the job-search and hiring processes (Bayer et al., 2008; Rajkumar et al., 2022).

4 Earnings Consequences

This section analyzes the returns to working for a parent's employer. I begin by defining the target estimand and characterizing the empirical challenges, then present the research design and estimates.

4.1 Target Estimand

Let the log earnings of individual i employed at firm $j(i)$ be $y_i = h_i + f_{j(i)} + m_{ij(i)} + u_i$, where h_i , $f_{j(i)}$, $m_{ij(i)}$, and u_i capture worker-specific, firm-specific, match-specific, and idiosyncratic factors, respectively.

Using the potential outcomes framework, let $j^1(i)$ denote the employer of individual i if they work for their parent's employer, and let $j^0(i)$ denote their outside option, defined as the employer they would work for otherwise. Potential earnings under treatment status $d \in \{0, 1\}$ are given by $y_i^d = y_{ij^d(i)}$. Observed earnings can therefore be written as

$$y_i = y_i^0 + D_i \beta_i, \quad (1)$$

where D_i is an indicator for whether individual i works for their parent's employer, and $\beta_i = y_i^1 - y_i^0 = (f_{j^1(i)} + m_{ij^1(i)}) - (f_{j^0(i)} + m_{ij^0(i)})$ is the treatment effect of working for a parent's employer.

This framework highlights two channels through which working for a parent's employer could impact earnings. First, the parent's employer might offer higher or lower pay to all workers:

$f_{j^1(i)} \neq f_{j^0(i)}$. This could reflect differences in productivity across firms or compensating differentials (Abowd et al., 1999; Sorkin, 2018). Second, firms may pay the children of current employees differently than otherwise similar workers: $m_{ij^1(i)} \neq m_{ij^0(i)}$. This could reflect reduced information asymmetries between the employer and job seeker or productivity gains from working alongside a parent (Montgomery, 1991; Heath, 2018).

The target estimand is the average treatment effect on the treated (ATT):⁹

$$\mathbb{E}[\beta_i | D_i = 1]. \quad (2)$$

The ATT answers the following question: Among individuals who work for their parent's employer, how much more do they earn there relative to their outside option?

The sign and magnitude of the ATT are theoretically ambiguous. The earnings gains could be positive if parental connections provide access to jobs that pay higher wages than the child's outside option. Alternatively, if individuals prefer to work at their parent's employer, they may accept a job there that pays less than their outside option.

4.2 Evidence of Negative Selection

A possible empirical strategy would compare the earnings of those who work for their parent's employer and those who do not. Such a comparison identifies the ATT plus a selection term:

$$\mathbb{E}[y_i | D_i = 1] - \mathbb{E}[y_i | D_i = 0] = \underbrace{\mathbb{E}[\beta_i | D_i = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[y_i^0 | D_i = 1] - \mathbb{E}[y_i^0 | D_i = 0]}_{\text{Selection}}. \quad (3)$$

I argue that, even with a rich set of covariates, as in Kramarz and Skans (2014), such a comparison is likely to result in downward-biased estimates due to negative selection.

To make this point, I show that even when comparing siblings within the same family, those who work for a parent's employer are less educated (i.e., they are negatively selected). I limit the sample to cases where both an individual and at least one sibling respond to the American Community

⁹Unless otherwise stated, $\mathbb{E}[\cdot]$ denotes an expectation over individuals i .

Survey (ACS) after age 25. Column 1 of Table A.2 presents estimates from a regression of an indicator for whether the individual's first job was at a parent's employer on educational attainment (measured in the ACS), controlling for household, first-job-quarter, and race-by-sex-by-birth-year fixed effects. Siblings with higher levels of education are significantly less likely to work for a parent's employer: compared to those without a high school degree, those with a bachelor's degree or higher are 3.1 percentage points less likely to work for a parent's employer. This is a large difference, considering that only 3.8 percent of individuals with a bachelor's degree work for a parent's employer. In other words, those without a college degree are almost twice as likely to work for their parent's employer as their college-educated siblings.

Column 2 of Table A.2 regresses initial log earnings on an indicator for whether the individual works for their parent's employer and the same fixed effects as Column 1. Siblings who work for a parent's employer at their first job earn 3.8 log points more. Column 3 adds controls for educational attainment and the coefficient rises to 4.1 log points. The increase in the R-squared between Columns 2 and 3 is modest and raises the possibility of selection on unobserved characteristics. Under the assumption that the full set of observed and unobserved characteristics perfectly predicts the outcome, the restricted estimator developed by Oster (2019) suggests that the change in coefficient and R-squared values between Columns 2 and 3 implies a true treatment effect of 19 log points. Thus, these estimates may be severely downward biased by selection on unobserved characteristics.

Even among siblings within the same household, those who work for a parent's employer are negatively selected. Any empirical strategy must account for selection.

4.3 Instrumental Variables Estimator

An ideal experiment to estimate the causal effect of working for a parent's employer would randomly prohibit some firms from hiring the children of current employees. Random assignment across firms would serve as an instrument, generating variation in access to jobs at a parent's employer that is orthogonal to other determinants of child outcomes. With perfect compliance, this design would identify the ATT. This design allows for the possibility that some children would choose not to work for their parent's firm and some firms would choose not to offer them a job (i.e., there exist

never-takers).

I approximate this ideal experiment using the hiring rate at the parent’s firm as an instrument. The intuition is straightforward: when a firm is hiring less, it is less likely to extend a job offer to the child of an existing employee. Two main identification concerns arise. First, hiring rates may be correlated with broader local labor market conditions, which could independently affect young workers’ outcomes. Second, parents employed at firms with higher hiring rates may differ systematically from those at firms with lower hiring rates. For example, larger firms may have lower hiring rates and employ workers with higher levels of education.

To address these concerns, I isolate variation in hiring rates that is both transitory and idiosyncratic to the parent’s employer by estimating the following two-stage least squares specification:

$$\begin{aligned} \text{First stage: } D_i &= \pi^1 + \gamma Z_{p(i)t-1} + \delta_{p(i)}^1 + \lambda_{l(p(i),t)}^1 + \phi^1 x_{it-1} + u_i \\ \text{Second stage: } y_i &= \pi^2 + \beta D_i + \delta_{p(i)}^2 + \lambda_{l(p(i),t)}^2 + \phi^2 x_{it-1} + v_i, \end{aligned} \tag{4}$$

where individual i starts their first stable job in quarter t . The main outcome, y_i , is initial log earnings at the first job. The endogenous variable, D_i , indicates whether i works for their parent’s employer at their first stable job. If i has two parents, I assign the parent with the higher average earnings between $t - 4$ and $t - 1$. The specification controls for fixed effects for the parent’s employer, $\delta_{p(i)}$; fixed effects for the local labor market of the parent’s employer, $\lambda_{l(p(i),t)}$, defined as the interaction between the commuting zone, two-digit industry, and quarter; and log parental earnings, x_{it-1} , measured as the average between quarters $t - 4$ and $t - 1$. Standard errors are clustered at the level of the parent’s employer, and the sample is restricted to parents with at least one year of tenure and excludes singleton observations.¹⁰

I instrument for D_i with $Z_{p(i)t-1}$, which is the average quarterly hiring rate at the parent’s employer measured in the year before the first job begins (i.e., quarters $t - 4$ to $t - 1$).¹¹ Using the

¹⁰Singletons—observations that have a unique value of a fixed effect—are excluded because they do not contribute to identification and bias the standard error estimates.

¹¹Following the Quarterly Workforce Indicators, I measure the End-of-Quarter Hiring Rate, which is defined as the number of new hires who remain with the firm for at least one more quarter, divided by the average of the total employment at the beginning and end of the quarter.

lagged hiring rate ensures that joining a parent's employer does not mechanically affect the hiring rate; see Appendix C for details.

Identification comes from changes over time in hiring at the parent's employer relative to other firms in the same local labor market. The design compares children whose parents work for the same firm but who enter the labor market at different times. The first stage asks whether children are less likely to start their careers at their parent's employer when that firm is hiring less than usual, relative both to its own hiring in other periods and to hiring at other firms in the same commuting zone and industry.

Panels A and B of Figure 2 demonstrate that the empirical strategy exploits transitory fluctuations in hiring conditions. Each point represents an estimate from a separate regression that varies Equation 4 by measuring the hiring rate at different points in time. The blue diamond in Panel A presents first-stage estimates, showing that the hiring rate at the parent's employer in the year before the child starts their first job strongly predicts whether they will work there. A 10 percentage point increase in the hiring rate leads to a 2 percentage point increase in the probability of working for the parent's employer. In contrast, when the hiring rate is measured three years earlier, the first-stage coefficient drops to 0.01—20 times smaller than the baseline estimate. Panel B shows a similar pattern of decay in the reduced form. These results indicate that young workers' outcomes depend on hiring conditions just before they enter the labor market—not conditions further in the past. This illustrates the role of the parental employer fixed effects, which account for time-invariant differences across firms.

Panels C and D of Figure 2 show that the empirical strategy exploits variation in hiring conditions that is idiosyncratic to the parent's employer and orthogonal to broader labor market trends. Each point represents an estimate from a separate regression, with Panels C and D displaying first-stage and reduced-form estimates, respectively. The blue diamonds correspond to the main specification from Equation 4, while the red circles represent placebo specifications, where all variables related to the parent's employer are replaced with those of a firm that is observably similar to the parent's employer. Specifically, I sample ten firms that are in the same commuting zone, two-digit industry,

and size class (above or below 500 employees) as the parent's employer. Hiring rates at placebo firms are largely unrelated to young workers' outcomes, highlighting the role of the local labor market fixed effects. Prior research finds that earnings are affected by aggregate economic conditions at the time an individual starts a new job or enters the labor market (Wachter, 2020). By controlling for local labor market fixed effects, my analysis isolates the effect of working for a parent's employer from the broader impact of finding a first job in a strong labor market.

Figure A.7 reports balance tests relating the children's baseline characteristics to hiring rates at the parents' employers. Baseline characteristics exclude variables directly controlled for in Equation 4 and include marital status of the parents, race, sex, year of birth, and a vector of characteristics of the Census tract in which the child lived in 2000. Hiring rates are strongly correlated with child characteristics in the raw data, but these correlations are eliminated after controlling for the covariates in the baseline specification. For instance, the coefficient on predicted initial earnings, constructed using all baseline characteristics, falls by more than three orders of magnitude and becomes statistically insignificant.

Three assumptions are needed to interpret estimates from Equation 4 as causal. The first is relevance: the hiring rate must affect the probability of working for a parent's employer. The second is monotonicity, which requires that the hiring rate weakly increase the probability of working at a parent's employer for all individuals. The third is exogeneity: conditional on the covariates, the hiring rate affects the earnings of the individual only through its effect on the probability of working for the parent's employer.

The exogeneity assumption has three key components. The firm-specific, transitory fluctuations in hiring at the parent's employer—i.e., the variation isolated by Equation 4—must be unrelated to (1) labor demand shocks at other firms, (2) changes in wages at the parent's employer, and (3) characteristics of the child—such as education or skills—that have an independent effect on earnings.

Under these assumptions, the instrumental variables estimator identifies a *local average treatment effect* (LATE), which is the average effect for the *compliers*—the population whose treatment status depends on the instrument.

4.4 Effect of Working for Parent’s Employer on Initial Earnings

Table 2 presents the main estimates from Equation 4. Columns 1 and 2 present the first-stage and reduced-form estimates, showing that a 10 percentage point increase in the hiring rate at the parent’s employer leads to a 2.2 percentage point increase in the probability that the individual works for their parent’s employer at their first stable job (the first-stage F-statistic is 6,993) and a 0.47 log point increase in initial earnings. Column 3 presents estimates from the second stage and shows that working for a parent’s employer increases initial earnings by 21 log points, or 24 percent ($24 = (e^{0.212} - 1) \times 100$). Column 4 presents the corresponding ordinary least squares estimate, which is likely substantially downward biased for the reasons discussed in Section 4.2.

The estimated earnings benefits are large but consistent with other evidence of the importance of place of work in determining earnings. For example, the estimated effect is about the same size as the union wage premium (Farber et al., 2021) and one standard deviation of the inter-industry wage premiums (Katz et al., 1989). For comparison, the college premium—the relative wage for college versus high-school educated workers—is 68 log points (Acemoglu and Autor, 2011).

Column 5 of Table 2 provides initial evidence on mechanisms by estimating the effect on the AKM fixed effect of the first employer.¹² Working for the parent’s employer leads individuals to work for firms that pay all workers 23 log points more, which is very similar to the effect on individual earnings in Column 3. I split the sample by the median pay premium of the parent’s employer, and find that working for the parent’s employer leads to a 25 (s.e.=2.6) and 10 (s.e.=2.7) log point increase in initial earnings for individuals whose parents are employed by high- and low-paying firms, respectively. These results suggest that the earnings gains are driven by parents providing access to higher-paying firms. Section 4.6 explores mechanisms in more detail.

4.5 Assessing the Plausibility of the Identifying Assumptions

Table 3 presents four sets of analyses that assess the three main components of the exogeneity assumption, defined in Section 4.3.

¹²I estimate the AKM firm fixed effect using code adapted from Crane et al. (2022) and based on national data that excludes the young workers in my sample. See Appendix B.3 for details.

The first concern is that hiring conditions at the parent's employer may be correlated with broader labor demand shocks. For instance, if parents transmit specific skills to their children and the parent's employer hires more when demand for those skills is high, the independence assumption would be violated, leading to upward bias in the estimated effects. To address this, the main specification controls for local labor market fixed effects, defined as the interaction of commuting zone, industry, and quarter. These controls absorb time-varying shocks to labor demand within narrowly defined markets. However, they do not account for the possibility that firms competing to hire the same workers might be in different industries or regions.

To assess this concern, I use hiring conditions at the parent's future employer as a proxy for such unobserved labor demand shocks. The intuition is that skill transmission occurs throughout childhood, so the types of skills parents pass on are likely to be valued by both the parent's current and future employers. Since children are less likely to have connections to the parent's future firm, this strategy isolates variation in demand for particular skills from the effects of parental connections.

I estimate the following equation:

$$y_i = \pi + \gamma Z_{p(i)t-1} + \delta_{p(i)} + \lambda_{l(p(i),t)} + \theta \tilde{Z}_{f(i)t-1} + \tilde{\delta}_{f(i)} + \tilde{\lambda}_{l(f(i),t)} + \phi x_{it-1} + \eta_i, \quad (5)$$

where $f(i)$ is the future employer of i 's parent. Equation 5 is a modification of the main specification (Equation 4) that includes controls for the hiring rate at the parent's future employer, $\tilde{Z}_{f(i)t-1}$, as well as fixed effects for the parent's future employer, $\tilde{\delta}_{f(i)}$, and its local labor market, $\tilde{\lambda}_{l(f(i),t)}$. I define the future employer as the parent's subsequent employer and the sample is limited to children whose parents have a future employer.

Column 1 of Table 3 presents estimates from Equation 5 and shows that the initial earnings of the young worker are strongly correlated with the hiring rate at the parent's current employer but unrelated to the hiring rate at the parent's future employer. Columns 2 and 3 present second-stage estimates that omit and include controls for the hiring rate at the parent's future employer, respectively.¹³ The estimated effect of working for a parent's employer is unaffected by controlling for hiring at the

¹³Both specifications control for fixed effects for the parent's future employer and its local labor market.

parent’s future employer. These results argue against bias from correlated labor demand shocks.

The second identification concern is that hiring activity at the parent’s employer may be correlated with changes in firm-level wages. While fixed effects for the parent’s employer absorb time-invariant differences across firms, they do not directly account for the possibility that expanding firms may both hire more workers and raise wages, violating the exclusion restriction and biasing the estimated effect upward. This concern is partly mitigated by the transitory nature of the identifying variation (see Figure 2), as firms are less likely to adjust wage policies in response to short-term hiring fluctuations. Furthermore, the rent-sharing literature finds that productivity shocks lead to only modest wage increases (Card et al., 2018) and Mueller et al. (2023) provide evidence that “firms’ wage policies do not constitute an important margin of recruiting effort,” a pattern that is especially plausible for young workers in entry-level positions.

To assess this concern more directly, Column 4 of Table 3 presents estimates from a variant of Equation 4 that includes two additional firm-level controls: the employment growth rate and the change in average earnings for incumbent workers, both measured just before the child starts their first job.¹⁴ The estimated effect of working for a parent’s employer is slightly larger in this specification—26 log points compared to 21 log points in the baseline—providing no evidence that the main results are driven by expanding firms offering higher wages.¹⁵

The third identification concern is that hiring conditions might be correlated with characteristics of the child—such as education or skills—that independently affect earnings. One way in which this could happen is if parents dynamically sort across firms in a way that produces a correlation between the hiring rate and the characteristics of the employees. For example, if firms respond to negative shocks by reducing hiring and retaining only higher-skilled workers, parents employed during periods of low hiring will tend to be more skilled.

I address the concern that parents dynamically sort across firms using a sibling comparison. Columns 5 and 6 of Table 3 restrict the sample to individuals who have at least one sibling. Column

¹⁴The employment growth rate is calculated as the change in employment between quarters $t - 5$ and $t - 1$, capturing annual growth that is unaffected by seasonal variation. Changes in average earnings are computed among workers continuously employed from $t - 8$ to $t - 1$, comparing mean quarterly earnings in periods $t - 8$ to $t - 5$ and $t - 4$ to $t - 1$.

¹⁵Results are similar when controlling for average earnings of other new hires: $\hat{\beta} = 0.262$, s.e. = 0.019.

5 includes household fixed effects, and Column 6 presents estimates from the baseline specification. I find similar earnings gains when controlling for fixed effects for the household and the parent's employer, which rules out all confounders, observed or unobserved, that are common to siblings within the same family.

I use education data from the American Community Survey (ACS) to directly test for sorting on observable characteristics. I restrict the sample to individuals who respond to the ACS after age 25—after most educational and early labor market decisions have been made. Columns 7 and 8 of Table 3 show that the estimated earnings effects are nearly identical with and without controls for the child's educational attainment.¹⁶ Column 9 augments the main specification to include the set of baseline characteristics included in the balance tests in Figure A.7. The main results are largely unaffected by the inclusion of these covariates. Thus, observable characteristics of the child appear to be unrelated to hiring conditions at the parent's employer.

Another concern is that parental connections could influence when, or even whether, a child enters the labor market. I present three pieces of evidence to address this concern. First, measuring the hiring rate independently of the actual timing of labor market entry yields similar results. Column 10 of Table 3 presents estimates from a specification in which the instrument is the average quarterly hiring rate at the parent's employer during the year the child turns 18.¹⁷ This alternative instrument yields similar estimates of the earnings gains: 18 log points. Second, hiring conditions are unrelated to extensive-margin responses. Columns 11 and 12 extend the sample to include all young workers, including those who never find a first stable job. In Column 11 the outcome is initial earnings, which is coded as zero for those who never find a first job. I estimate the second stage using a Poisson model (Chen and Roth, 2024) and find similar point estimates, although the standard errors are larger and the estimates are not statistically different from the estimates in Column 10 or from zero. Column 12 presents reduced-form estimates and shows that the hiring rate at the parent's employer

¹⁶I also find no effect when education is the outcome: working for a parent's employer leads to a statistically insignificant 3 percentage point reduction (p-value=0.87) in the probability of obtaining a bachelor's degree.

¹⁷The local labor market fixed effects are also defined based on the parent's employer when the child is 18. The sample includes individuals whose parent is employed at the same firm between the ages of 18 and 22, which is the five-year window in which the most children enter the labor market.

in the year the child turns 18 is unrelated to whether the individual ever finds a first job. Third, I find that 99.6 percent (s.e.=5.7) of the untreated compliers find a first job.¹⁸

As a final test of the identification assumptions, I examine whether the results are stronger in industries where hiring through social networks is more common. Motivated by Figure A.6—which documents substantial variation in the use of informal connections across industries—I explore treatment-effect heterogeneity by industry. I calculate the share of individuals whose first job is at a parent’s employer in each three-digit industry and group industries into deciles based on this share.

Panel A of Figure A.8 presents first-stage estimates from a variant of Equation 4 that interacts the hiring rate instrument with industry decile. The results show that, in industries where family-based hiring is more common, increases in the hiring rate lead to larger increases in the probability that a child works for their parent’s employer. The first-stage estimates are positive for all deciles, supporting the monotonicity assumption.

Panel B of Figure A.8 presents second-stage estimates of the effect of working for a parent’s employer on initial earnings by industry decile. The results show no systematic relationship between the size of the earnings gain and the prevalence of family-based hiring. This pattern is reassuring, as it helps rule out violations of the independence assumption that apply to industries in which family-based hiring is less common. Intuitively, if the reduced-form relationship between earnings and the hiring rate were attributable to factors unrelated to working for a parent’s employer—such as aggregate labor market conditions—then we would expect to see larger second-stage estimates in industries where the use of parental connections is rare.

As an alternative test of the validity of the estimates, I use a distinct research design to estimate the returns to parental connections. I focus on young workers who are displaced from rapidly contracting firms and compare displaced co-workers who subsequently find a job at their parent’s employer with those who do not, adjusting for observable worker characteristics. The key identification assumption is that displaced workers who join a parent’s employer would otherwise have had similar earnings

¹⁸I estimate the untreated complier mean by estimating the two-stage least squares specification where the outcome is the interaction between an indicator for ever finding a first job and one minus the treatment, and the treatment variable is replaced by one minus treatment. See Angrist and Imbens (1995) and Garin et al. (2025) for a discussion of estimating complier means with a continuous instrument.

trajectories to observably similar co-workers who do not. This approach yields similar estimates: working for a parent’s employer increases post-displacement earnings by 21 log points and firm pay premiums by 19 log points. Appendix D discusses the design and results in detail.

The displaced-worker analysis is reassuring because it relies on a distinct set of identifying assumptions from the main design. Although it identifies a different parameter—the effect of parental connections on recovery from job loss rather than access to a first job—the similar estimates provide a useful validation of the main results.

4.6 Mechanisms and Other Results

Parents provide access to higher-paying firms partly because those firms are in higher-paying industries. Column 1 of Table 4 shows that working for a parent’s employer increases the two-digit industry-level pay premium of the child’s first employer by 14 log points.¹⁹ This accounts for 67 percent of the increase in individual-level earnings. Columns 2 through 4 show that working for a parent’s employer reduces the likelihood of working in the unskilled service sector by 42 percentage points and increases the likelihood of working in the blue-collar sector by 37 percentage points.

The shift from the unskilled service sector to the blue-collar sector is consistent with the evidence of negative selection discussed in Section 4.2. Estimating a variant of Equation 4 where the outcome is $y_i \times (1 - D_i)$ and endogenous regressor is $(1 - D_i)$ instead of D_i , implies that the mean initial log earnings among untreated compliers is 8.58, which is 19 log points less than mean earnings among those who do not work for a parent’s employer. Intuitively, young workers with low-paying outside options are more likely to rely on parental connections to find work.

Columns 5 and 6 of Table 4 illustrate that working for the employer of a parent leads individuals to start their careers on a higher rung of the firm job ladder as defined by average earnings and the proportion of hires made through poaching flows.²⁰ Columns 7 and 8 show that individuals who work for their parent’s employer end up at smaller and younger firms, possibly because these firms

¹⁹Industry pay premiums are estimated using national data excluding the young workers in my sample, by regressing log earnings on individual and industry fixed effects. See Section B.3 for details.

²⁰The outcomes in Columns 5 and 6 correspond to the rank of time-invariant characteristics of the first employer relative to the national distribution of firms.

tend to use less formal hiring practices.

The outcome in Column 9 of Table 4 is the child's earnings rank within their first employer. The effect is negative, implying that, while parents provide access to higher-paying firms, they do not provide access to relatively high-paying jobs within firms.

Working for a parent's employer increases tenure at the first employer. Column 1 of Table 5 indicates that working for a parent's employer increases the probability of remaining at the first employer for at least three years by 20 percentage points. Columns 2 and 3 illustrate that this effect reflects fewer job-to-job transitions, rather than fewer transitions from employment to nonemployment. If the outcomes in Columns 2 and 3 proxy quits and layoffs, respectively, then these results suggest that parental connections provide access to firms that are more desirable than the child's outside option.

Columns 4 through 6 of Table 5 illustrate that the earnings benefits are quite persistent. Working for a parent's employer increases annual earnings by \$6,683 in the first year after finding a first job. By the third year, those who started their careers at their parent's employer earn \$5,566 more.²¹ The outcome in Column 7 is the growth in earnings between the first and third year and shows that individuals who work for their parent's employer experience slower earnings growth. These effects are consistent with parents providing access to jobs on a higher rung of the firm job ladder. The initial gains are large but they slowly fade over time as other young workers move to higher-paying jobs and experience slightly stronger earnings growth.

4.7 Interpreting the Local Average Treatment Effect

Does the instrumental variables estimator identify effects for a narrow subgroup, or are the estimates broadly representative of all young workers who use parental connections?

The independence and monotonicity assumptions imply that the ATT can be written as a weighted average of treatment effects for compliers and always-takers. This implies that the LATE equals the ATT if there are no always-takers or if compliers and always-takers have the same average

²¹To facilitate a comparison with Table 2, Table A.5 presents analogous results from a Poisson model, which imply a 17 percent increase in earnings three years later.

treatment effects. See Appendix E for details.

I present three sets of estimates to investigate the relationship between the LATE and the ATT. First, I show that the instrument induces a substantial share of individuals into treatment. Moving from the 5th to the 95th percentiles of the hiring distribution increases the likelihood of working for a parent's employer by 2.8 percentage points (see Figure A.10). This is large considering that 6 percent of individuals work for a parent's employer within the estimation sample, suggesting many treated individuals are compliers.

Second, I compare the mean of log initial earnings in the treated complier sample and the full treated sample. Replacing the second-stage outcome in Equation 4 with $y_i \times D_i$ implies mean initial log earnings of 8.79 for treated compliers, which is only 5 log points greater than the full-treated-sample mean of 8.74. This shows that the compliers are similar to the general treated population in terms of potential earnings, y_i^1 .

Third, while there is evidence of selection on gains, the magnitude is modest. Following Cornelissen et al. (2016), I estimate marginal treatment effects (MTE) under the assumptions of full independence and linear separability. See Appendix F for estimation details and Figure A.11 for the estimated MTE curve. The MTE is 29 log points at the mean value of the estimated propensity score within the first decile and falls slightly to 25 log points at the ninth decile. The MTE drops more steeply to 19 log points in the tenth decile. These results imply selection on gains: individuals who benefit more from working for a parent's employer are less resistant to doing so. However, the range of the MTE is modest over most of the identified support of the estimated propensity score.

The above empirical results suggest that the LATE may be a reasonable approximation of the ATT in this setting. But why might selection on gains be modest in this setting? One possibility is that accepting a job at a parent's employer is driven more by the amenity value of working with a parent than by expected earnings gains. A second possibility arises from the fact that treatment in this context depends on two decisions: the parent's employer must make an offer, and the young worker must accept it. If the employer's offer decision is unrelated to worker-side determinants of acceptance, then the LATE does not vary with the instruments and equals the ATT even when there

is selection on gains (see Appendix E). This condition may hold if firms seek workers likely to stay with the firm, who may not be those with the highest earnings potential.²²

5 Intergenerational Persistence in Earnings

This section examines how parental connections shape the intergenerational persistence in earnings. Section 5.1 defines the estimand of interest and the empirical approach. Section 5.2 presents the results. Section 5.3 interprets the findings using a stylized model of intergenerational mobility.

5.1 Target Estimand

My objective is to estimate how the intergenerational elasticity of earnings (IGE)—a common measure of the intergenerational persistence in earnings—would change if no one worked for their parent’s employer. The IGE is the coefficient obtained from regressing the log earnings of the child, y_i , on the log earnings of their parents, y_p , and is denoted

$$\rho(y_i, y_p) \equiv \frac{\text{cov}(y_i, y_p)}{\text{var}(y_p)}. \quad (6)$$

The estimand of interest is the difference between the observed IGE and the counterfactual IGE in which no one worked for their parent’s employer. Combining equations 1 and 6 yields the following expression for this estimand:

$$\rho(y_i, y_p) - \rho(y_i^0, y_p) = \frac{\text{cov}(D_i \beta_i, y_p)}{\text{var}(y_p)}. \quad (7)$$

Equation 7 shows that the intergenerational transmission of employers will increase the intergenerational persistence in earnings if the benefits, $D_i \beta_i$, are increasing in parental earnings. This occurs if individuals with higher-income parents are more likely to work for a parent’s employer and experience larger earnings gains when they do.

²²Consistent with this view, Burks et al. (2015) show that referred workers have similar measured skills to non-referred workers but are substantially less likely to quit.

Adding and subtracting the means of D_i and β_i yields the following decomposition:

$$\frac{\text{cov}(D_i\beta_i, y_p)}{\text{var}(y_p)} = \bar{D} \frac{\text{cov}(\beta_i, y_p)}{\text{var}(y_p)} + \bar{\beta} \frac{\text{cov}(D_i, y_p)}{\text{var}(y_p)} + \frac{\text{cov}((D_i - \bar{D})(\beta_i - \bar{\beta}), y_p)}{\text{var}(y_p)}, \quad (8)$$

where $\bar{x} \equiv \mathbb{E}[x]$. The three terms capture the contributions of heterogeneous returns to connections, differential access to connections, and the interaction between the two.

To estimate the covariance term, I rely on the following decomposition:

$$\text{cov}(D_i\beta_i, y_p) = \mathbb{E}_{y_p} \left[\mathbb{E}[D_i|y_p] \mathbb{E}[\beta_i|y_p, D_i = 1] y_p \right] - \mathbb{E}[D_i] \mathbb{E}[\beta_i|D_i = 1] \mathbb{E}[y_p], \quad (9)$$

where $\mathbb{E}_{y_p}[\cdot]$ denotes an expectation over y_p . I estimate the key inputs, $\mathbb{E}[D_i|y_p]$ and $\mathbb{E}[\beta_i|y_p, D_i = 1]$, using the approaches described in Sections 3 and 4. In practice, I approximate Equation 9 by conditioning on quantiles rather than exact values of parental earnings. See Appendix G for details.

A key advantage of this methodology is that it accounts for nonrandom selection into a parent's employer. This contrasts with common approaches that assess changes in the IGE after adding controls. For example, Corak and Piraino (2011) and Stinson and Wignall (2018) compare baseline intergenerational earnings regressions to specifications that control for whether an individual works for their parent's employer.

5.2 Counterfactual Estimates

Individuals with higher-earning parents are more likely to work for their parent's employer. Figure 3 presents the proportion of individuals who work for a parent's employer for each percentile of the parental earnings distribution. Only 2 percent of individuals with parents at the bottom percentile of the earnings distribution work for a parent's employer. In contrast, 7 percent of individuals with parents at the top percentile of the earnings distribution work for a parent's employer. I find similar disparities using longer-run measures. Figure A.12 shows that 31 percent of individuals whose parents are in the top decile of the earnings distribution work for a parent's employer at some point between ages of 16 and 30, compared to 25 percent among those in the bottom decile.

Individuals with higher-earning parents may be more likely to work for their parent's employer

because their parents are more likely to be employed and hold a position of authority within the firm. The percentage of individuals who have an employed parent when they find their first job rises steeply from 42 percent to 63 percent between the 1st and 20th percentiles of the parental earnings distribution and eventually plateaus at 85 percent. The percentage of individuals who have a parent who is a top earner at their firm rises gradually from 3 to 14 percent between the 1st and 90th percentiles of the parental earnings distribution and then rises steeply to 33 percent in the top percentile.²³ Figure A.14 reproduces Figure 3 but for the sample of individuals whose parents are employed and not in the top percentile of the within-firm earnings distribution. Implementing these sample restrictions eliminates the positive gradient with parental earnings.

Individuals with higher-earning parents experience larger earnings gains from working for a parent's employer. Figure 4 presents estimates from the main instrumental variables specification estimated on five distinct samples defined by the quintile of the parental earnings distribution. Among those in the bottom quintile, working for a parent's employer leads to a statistically insignificant 10 log point increase in initial earnings. In contrast, working with a parent in the fourth and fifth quintiles of the earnings distribution leads to a 28 and 16 log point increase in earnings, respectively. The estimated effects on the AKM firm effect follow a similar pattern. Panels A and B of Table A.6 present estimates by the quintile of parental earnings using the instrumental variables estimator and the mass-layoff estimator, respectively. The two empirical strategies yield quantitatively similar results: individuals with higher-earning parents experience larger gains in both individual earnings and the AKM firm effect.

Table 2 shows that working for the employer of a parent who is the primary earner in the household leads to a 21.2 log point increase in initial earnings. Repeating this analysis for the parent who is the secondary earner yields a similar earnings gain of 20.7 log points ($p\text{-value} < 0.001$). Thus, for the counterfactual exercise, I assume that the earnings gains from using parental connections are the same for both parents.

What are the implications for the IGE? Column 1 of Table 6 presents estimates of the IGE,

²³Figure A.13 presents these results in detail by plotting the proportion of individuals with parents who are employed and that are top earners within their employer against the percentile of parental earnings.

which is the coefficient from a regression of the initial log earnings of the child at their first job on the log earnings of their parents. The estimated IGE of 0.136 is substantially lower than other estimates in the literature (Black and Devereux, 2010).²⁴ To understand why, Table A.7 presents alternative estimates of the IGE. The IGE is 0.482 when child earnings are measured between ages 29 and 31 and one is added to earnings to include individuals with zero earnings. The IGE drops to 0.162 if I then omit children whose average quarterly earnings between ages 29 and 31 do not exceed \$3,300 (the same restriction used to define the first stable job).²⁵ These patterns demonstrate that the lower IGE observed in my sample is primarily because the sample is restricted to children who are employed. It is worth emphasizing that estimates of the intergenerational persistence in earnings typically use long-run measures of earnings for both parents and children, whereas I focus on children's initial labor market outcomes.

Column 2 of Table 6 shows that the IGE would be 12 percent lower if no one worked for a parent's employer. This estimate is constructed using the methodology described in Equation 7 in combination with the descriptive statistics in Figure 3 and the instrumental variables estimates in Figure 4. Standard errors are calculated using the delta method and take into account the uncertainty in the estimates of the earnings consequences. These findings follow intuitively from the fact that young workers with higher-earning parents benefit more from parental connections because they are more likely to work for a parent's employer and experience larger earnings gains when they do.²⁶

These findings are consistent with the evidence that parental connections help individuals with low-paying outside options obtain higher-paying blue-collar jobs. As discussed above, Table A.2 shows negative selection *within* families, implying that individuals who use parental connections appear to have weaker outside options for individual-level rather than family-level reasons. Indeed, within each parental earnings quintile, untreated compliers earn substantially less than those who do

²⁴Figure A.15 presents a visual representation of the IGE by plotting the average log earnings at the first job against the average log earnings of the parents for each percentile of the parental earnings distribution. The flatter slope at the bottom of the parental earnings distribution is likely attributable to the fact that, by construction, everyone in my sample has a stable job.

²⁵Column 4 of Table A.7 shows that I find a similar IGE if I measure parental earnings as the average earnings in the years when the child was between the ages of 16 and 20.

²⁶Figure A.15 presents a visualization of these results and compares the observed IGE to the counterfactual IGE.

not work for a parent’s employer (Figure A.16). Moreover, although children from higher-income families are less likely to work in blue-collar industries overall, this pattern reverses among those who use parental connections: among those who work for a parent’s employer, individuals with parents in the top earnings quintile are 14 percentage points more likely to work in blue-collar industries than those with parents in the bottom quintile.

Using the decomposition in Equation 8, Columns 3 and 4 of Table 6 show counterfactual estimates that hold either the earnings effects or the likelihood of using connections fixed at their means. Differential access accounts for 20 percent ($0.20 = 2.3/11.5$) of the total effect, while heterogeneity in returns accounts for 17 percent ($0.17 = 1.9/11.5$). The remaining 63 percent is due to the interaction between access and returns.

The conclusions are robust to conservative adjustments for the possibility that some individuals would have worked for these firms absent parental connections. The analysis of the parent’s future employers in Figure 1 demonstrated that up to 20 percent of individuals who work for a parent’s employer would have obtained the job absent parental connections. Column 5 of Table 6 shows that, if I adjust the probability of working for a parent’s employer accordingly—i.e., replace $\mathbb{E}[D_i]$ with $0.8 \times \mathbb{E}[D_i]$ —the counterfactual IGE would be 9.2 percent lower, instead of 11.5 percent.

Column 6 of Table 6 shows that I arrive at similar conclusions if I instead use estimates of the earnings consequences based on the mass-layoff analysis presented in Panel B of Table A.6: the IGE would be 10.1 percent lower.

Parental connections also amplify the initial gender pay gap. At their first job, young men earn 7 log points more than young women, on average. Table A.8 presents the estimated earnings gains from working for a parent’s employer based on the instrumental variables strategy for four samples defined by the sex of the child and the parent. Working for the father’s employer leads to a 28 and 23 log point increase in initial earnings for daughters and sons, respectively. Working for a mother’s employer leads to an 18 and 30 log point earnings gain for daughters and sons, respectively. Furthermore, sons are 2.6 times as likely to work for their father’s employer, while daughters are 1.3 times as likely to work for their mother’s employer. These estimates imply that the initial gender pay

gap would be 9 percent lower if no one worked for a parent’s employer.²⁷

I further disaggregate the results by sex, race, and ethnicity. Figure A.17 shows that for daughters, there are small differences across racial groups in the propensity to work for a parent’s employer conditional on parental earnings. In contrast, Black sons are significantly less likely to work for a parent’s employer relative to White sons whose parents are in the same percentile of the earnings distribution. This aligns with prior work by Chetty et al. (2020), who find that, conditional on parental income, Black males have lower expected income compared to White males. Figure A.18 replicates this finding and shows a conditional Black-White earnings gap of 8 log points in my sample. I calculate the counterfactual earnings for both groups—by subtracting $\mathbb{E}[D_i | y_p] \mathbb{E}[\beta_i | y_p, D_i = 1]$ from the average log earnings—and find that this conditional Black-White earnings gap would be 4 percent smaller if no one worked for their parent’s employer.²⁸

5.3 Stylized Model

I develop a stylized model to interpret the main empirical findings. I summarize the key insights here and provide the details in Appendix H.

Following the literature, I consider a model in which earnings depend on human capital, and children with higher-income families tend to have higher levels of human capital. I depart from existing models of intergenerational mobility by allowing earnings to also depend on a firm-level pay premium. Workers receive job offers through formal job search, and those with higher human capital tend to receive offers from firms with higher pay premiums. A parent’s employer may also make a job offer to the child. This offer decision depends on the human capital of both the child and the parent. The child will accept the offer if the earnings gain is sufficiently large, which occurs when the parent’s firm has a higher pay premium than the child’s outside option.²⁹

The model yields two insights. First, parental connections can, in theory, either increase or

²⁷I calculate this statistic using the same underlying methodology described in Section 5.1. Specifically, $-0.09 = (0.28 \times 0.015 + 0.18 \times 0.034 - 0.23 \times 0.039 - 0.30 \times 0.026)/0.07$.

²⁸This counterfactual exercise uses the earnings estimates in Figure 4 because I do not have sufficient power to estimate heterogeneous effects by sex, race, and parental earnings.

²⁹Magruder (2010) and Corak and Piraino (2010) develop models of mobility that incorporate parental contacts. Neither paper considers the role of firm pay premiums nor the endogenous use of social contacts.

reduce the intergenerational persistence in earnings. On the one hand, higher-earning parents may be better positioned to obtain high-paying job offers for their children. On the other hand, children from lower-income families have lower levels of human capital and may therefore benefit more from parental connections.

Second, parental investment decisions in children's human capital can amplify or dampen the direct effect of parental connections on the intergenerational persistence in earnings. Parents may invest more in their child's human capital if connections provide access to firms where skills are rewarded more highly. This force is likely strongest for higher-income parents, who can provide access to higher-paying firms. Alternatively, high-income parents may invest less if connections reduce the earnings penalty from having low human capital. The counterfactual exercise in this paper should therefore be interpreted as a partial-equilibrium analysis that holds children's human capital fixed.

6 Conclusion

This paper shows that the intergenerational persistence in earnings is attributable, in part, to parents using their connections to provide access to higher-paying firms. Existing research documents the widespread use of social contacts in the labor market but has less to say about their earnings consequences. I exploit transitory, firm-specific fluctuations in the availability of jobs at the parent's employer and estimate substantial earnings gains from finding a job through parental connections. Individuals with higher-earning parents are more likely to work for a parent's employer and experience larger earnings gains when they do; thus, connections at the parent's employer lead to a modest increase in the intergenerational persistence in earnings.

While connections to a parent's employer are clearly not the main determinant of the intergenerational persistence in earnings, individuals may find jobs through a wider set of social contacts, such as friends or extended family. Understanding how these broader connections shape intergenerational mobility should be a priority for future research. Furthermore, the analysis focuses on the first job, and future research should examine how unequal access to jobs through social connections shapes

labor market outcomes throughout the rest of the life cycle.

The results relate to the normative assessment of whether rates of intergenerational mobility are too low in the United States, an assessment that depends on whether the economic system is equitable and efficient. While equity depends on subjective moral values, a core ideal in the United States is that of equality of opportunity, which requires that an individual's success be a function of their hard work and ability rather than the circumstances into which they were born. Thus, from an equity standpoint, the findings raise concerns about the relatively low levels of intergenerational mobility in the United States. It is worth noting, however, that the primary beneficiaries of these parental connections are blue-collar workers, a group that has experienced declining labor market fortunes over the past few decades. The results do not speak directly to the implications for efficiency, and future research should aim to understand whether family connections lead to gains or losses in productivity.

The results also inform the positive assessment of what would be required to achieve equality of opportunity. One view is that economic rewards are determined by a worker's skills, which suggests that efforts to expand economic opportunity should aim only to equip everyone with the skills they need to succeed in the labor market. The results challenge this purely meritocratic view of the labor market, as individuals from high-income families earn more not only because they are more skilled, but also because their parents' connections provide access to high-paying firms. If connections in the labor market play a direct role in propagating intergenerational disadvantage, then achieving equality of opportunity in terms of education will not necessarily produce equality of opportunity. Rather, individuals from disadvantaged backgrounds may require additional support throughout their early careers. Gaining a better understanding of the mechanisms through which parents help their children find high-paying jobs may offer ideas for helping young workers who cannot rely on their parents' connections to navigate the labor market more successfully.

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Table 1: Summary Statistics

		Works for Parent's Employer	
	Full Sample (1)	No (2)	Yes (3)
A. Demographic Variables			
Age	21.5	21.6	20.6
Male	0.51	0.50	0.57
White, Non-Hispanic	0.71	0.71	0.74
Black, Non-Hispanic	0.10	0.10	0.07
Asian, Non-Hispanic	0.03	0.03	0.03
Hispanic	0.13	0.13	0.13
Single Parent	0.16	0.16	0.10
Parent is Top Earner in Firm	0.08	0.07	0.19
Parental Earnings (Thousands)	54.7	54.2	63.0
B. First Stable Job			
Annual Earnings (Thousands)	27.0	27.1	25.4
Stay for Three Years	0.40	0.39	0.56
Skilled Service Sector	0.36	0.35	0.38
Unskilled Service Sector	0.47	0.48	0.30
Blue-Collar Sector	0.17	0.17	0.32
Large Firm (Employees>500)	0.41	0.41	0.41
Urban	0.73	0.73	0.70
Observations (Millions)	25.86	24.51	1.35

Notes: This table presents the mean value of the variable defined by the row and for the sample defined by the column. All variables are measured in the year the child begins their first stable job. Column 1 presents results for the full sample. Columns 2 and 3 present results for the sample of children who do not and do work for a parent's employer at their first stable job, respectively. The variables in Panel A are demographic characteristics of the child and the variables in Panel B are characteristics of their first stable job.

Table 2: Effect of Working for Parent's Employer on Outcomes at First Job

	Works for Parent's Employer	Initial Log Earnings			AKM Firm Effect
	(1)	(2)	(3)	(4)	(5)
Hiring Rate at Parent's Employer	0.220*** (0.003)	0.047*** (0.004)			
Works for Parent's Employer			0.212*** (0.018)	0.004 (0.002)	0.229*** (0.014)
Description	First-Stage	Reduced- Form	Second- Stage	OLS	Second- Stage
First-Stage F-Statistic	6,993		6,993		6,922
Observations in Millions	17.6	17.6	17.6	17.6	17.49

Notes: Each column presents estimates from a separate regression corresponding to equation 4. Column 1 presents first-stage estimates, where the dependent variable is an indicator for whether the individual works for their parent's employer at their first stable job, and the main independent variable is the hiring rate at the parent's employer, measured in the four quarters before the first job begins. Column 2 presents reduced-form estimates, where the dependent variable is initial log earnings. Columns 3 and 5 present second-stage estimates, instrumenting for working for the parent's employer with the hiring rate; the dependent variables are initial log earnings and the AKM firm fixed effect of the first employer, respectively. Column 4 presents the corresponding ordinary least squares estimate. All regressions include fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, where the local labor market is defined by the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. Standard errors are clustered at the level of the parent's employer and are presented in parentheses.

*** $p \leq 0.001$, ** $p \leq 0.01$, * $p \leq 0.1$

Table 3: Assessing the Plausibility of the Identifying Assumptions

	Parent's Future Employer		Emp. Growth Controls		Household F.E.		Education Controls		Add. Controls		Extensive Margin	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Hiring Rate at Parent's Employer	0.030* (0.016)											
Hiring Rate at Parent's Future Employer	0.007 (0.008)		0.006 (0.008)									
Works for Parent's Employer		0.173* (0.092)	0.171* (0.093)	0.264*** (0.025)	0.167** (0.059)	0.159*** (0.037)	0.424* (0.195)	0.434* (0.191)	0.211*** (0.018)	0.179* (0.073)	0.147 (0.118)	
Hiring Rate at Parent's Employer, at Age 18												0.000 (0.003)
First-Stage F-Statistic		355.7 IV	354.3 IV	4,044 IV	642.2 IV	1,675 IV	99.99 IV	100.5 IV	6,973 IV	532.1 IV	605.4 IV	
Estimator	OLS											OLS
Observations in Millions	2.312	2.312	2.312	17.35	3.21	3.21	.44	.44	17.16	11.8	14.24	14.28

Notes: Each column presents estimates from a separate regression corresponding to instrumental variables (IV) estimator defined in equation 4. The outcome variable is defined by the column header. The specifications in Columns 1-9 measure the hiring rate at the parent's employer in the four quarters before the first job begins and control for fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, where the local labor market is defined by the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. Columns 1-3 additionally control for the hiring rate at the parent's future employer as well as fixed effects for the parent's future employer and the labor market in which that firm is located. Column 1 presents reduced-form estimates and Columns 2 and 3 present second-stage estimates from specifications with and without the control for the hiring rate at the parent's future employer. Column 4 controls for the employment growth rate and the change in average earnings for incumbent workers, both measured in the quarter before the child starts their first job. Column 5 includes household fixed effects and restricts the sample to individuals who have at least one sibling in the sample. Column 6 estimates the baseline specification on the same sample as Column 5. Columns 7 and 8 restrict the sample to individuals who respond to the ACS after age 25. Column 7 estimates the baseline specification while Column 8 adds fixed effects for educational attainment. Column 9 augments the baseline specification to also control for marital status of the parents; fixed effects for race, sex, and year of birth; and the vector of characteristics of the Census tract in which the child lived in 2000 that appears in Figure A.7. The specifications in Columns 10 through 12 measure the hiring rate at the parent's employer in the year the child turns 18 and control for fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, where the local labor market is defined by the interaction between commuting zone, industry, and the year in which the child turns 18; and log parental earnings, measured in the year the child turns 18. Column 10 presents second-stage estimates. Columns 11 and 12 expand the sample to include individuals who never find a first stable job. The outcome in Column 11 is earnings, where those who do not have a first job have zero earnings, and the specification is estimated via a Poisson regression using a control function approach. Column 12 presents reduced-form estimates. The reduced-form specifications in Columns 1 and 12 are estimated via ordinary least squares (OLS). Standard errors are clustered at the level of the parent's employer and are presented in parentheses.

*** p≤0.001, ** p≤0.01, * p≤0.1

Table 4: Effect of Working for the Parent's Employer on the Characteristics of the Child's First Employer

	Sector			Firm Ranking				Firm Age (8)	Log Firm Size (7)	Poaching Hires (6)	Average Earnings (5)	Blue-Collar (4)	Unskilled Services (3)	Skilled Services (2)	Industry Premium (1)	Within-Firm Earnings Rank (9)
	Industry Premium (1)	Skilled Services (2)	Unskilled Services (3)	Blue-Collar (4)	Average Earnings (5)	Poaching Hires (6)	Log Firm Size (7)									
Works for Parent's Employer	0.14*** (0.01)	0.05** (0.02)	-0.42*** (0.02)	0.37*** (0.02)	22.04*** (1.10)	7.72*** (0.99)	-1.21*** (0.10)	-3.15*** (0.53)								-6.45*** (1.06)
Mean	-0.13	0.37	0.47	0.16	43.80	54.50	5.66	23.62								39.30
Standard Deviation	0.16	0.48	0.50	0.37	27.00	23.30	2.47	12.78								24.70
First-Stage F-statistic	6,993	6,993	6,993	6,993	6,993	6,996	6,993	6,993								6,929
Observations (Millions)	17.60	17.60	17.60	17.60	17.60	17.60	17.60	17.60								17.14

Notes: This table presents second-stage estimates from the two-stage least squares specification defined in equation 4. The outcomes in Columns 1 through 8 are characteristics of the individual's first employer and are defined by the column header. The outcome in Column 9 is the individual's earnings rank within their first employer. All regressions include fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, where the local labor market is defined by the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. Standard errors are clustered at the level of the parent's employer and are presented in parentheses. The mean and standard deviation of the outcome variable correspond to the sample of individuals who do not work for their parent's employer.

*** p≤0.001, ** p≤0.01, * p≤0.1

Table 5: Effect of Working for the Parent's Employer on Job Mobility and Earnings Three Years After Entry

	Type of Job Transition			Annual Earnings After Entry			Three-Year
	Stay (1)	J2J (2)	J2N (3)	One Year (4)	Two Years (5)	Three Years (6)	Earnings Growth (7)
Works for Parent's Employer	0.20*** (0.02)	-0.15*** (0.02)	-0.04* (0.02)	6,683*** (646)	6,784*** (843)	5,566*** (954)	-0.07* (0.03)
Mean	0.36	0.41	0.23	26,700	26,800	28,100	1.06
Standard Deviation	0.48	0.49	0.42	15,900	20,300	22,700	0.63
First-Stage F-statistic	5,883	5,883	5,883	5,883	5,884	5,883	5,883
Observations (Millions)	14.98	14.98	14.98	14.98	14.98	14.98	14.98

Notes: This table presents second-stage estimates from the two-stage least squares specification defined in equation 4. The outcomes in Columns 1 through 3 are indicators for three different types of job transitions: (Stay) remained continuously employed at the first employer for three years, (J2J) changed jobs without experiencing a full quarter of nonemployment, and (J2N) separated from the first employer and experienced at least one full quarter of nonemployment. The outcomes in Columns 4 through 6 are annual earnings in the one to three years after the individual begins their first job. All regressions include fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, where the local labor market is defined by the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. Standard errors are clustered at the level of the parent's employer and are presented in parentheses. The mean and standard deviation of the outcome variable correspond to the sample of individuals who do not work for their parent's employer.

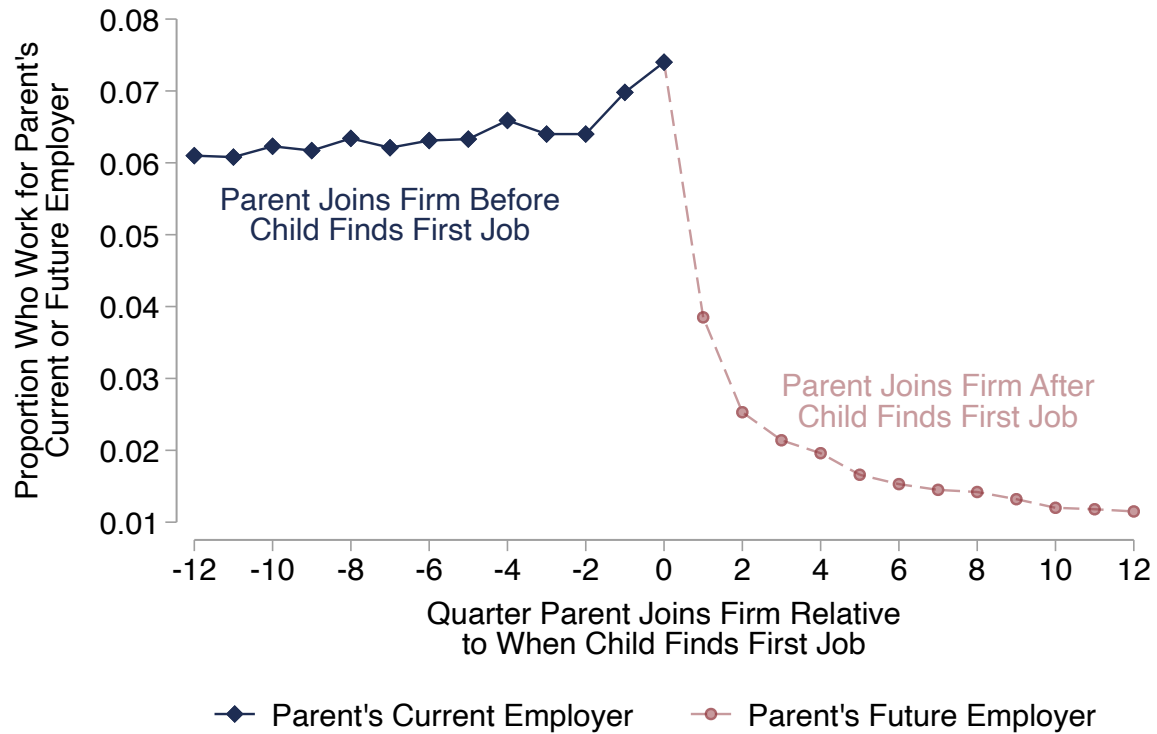
*** p≤0.001, ** p≤0.01, * p≤0.1

Table 6: Changes in Counterfactual Intergenerational Elasticity of Earnings

	Observed IGE	Percent Change in IGE if No One Worked for Parent's Employer				
	(1)	(2)	(3)	(4)	(5)	(6)
Estimate	0.136 (0.0001)	-11.5% (6.0)	-2.3% (0.2)	-1.9% (1.4)	-9.2% (4.8)	-10.1% (4.8)
Empirical Strategy Used to Estimate Earnings Gains		IV	IV	IV	IV	Mass Layoff
Assume No Heterogeneity In:						
Effect on Earnings			X			
Use of Connections				X		
Assume 20 Percent Would Have Obtained Job Absent Connections					X	

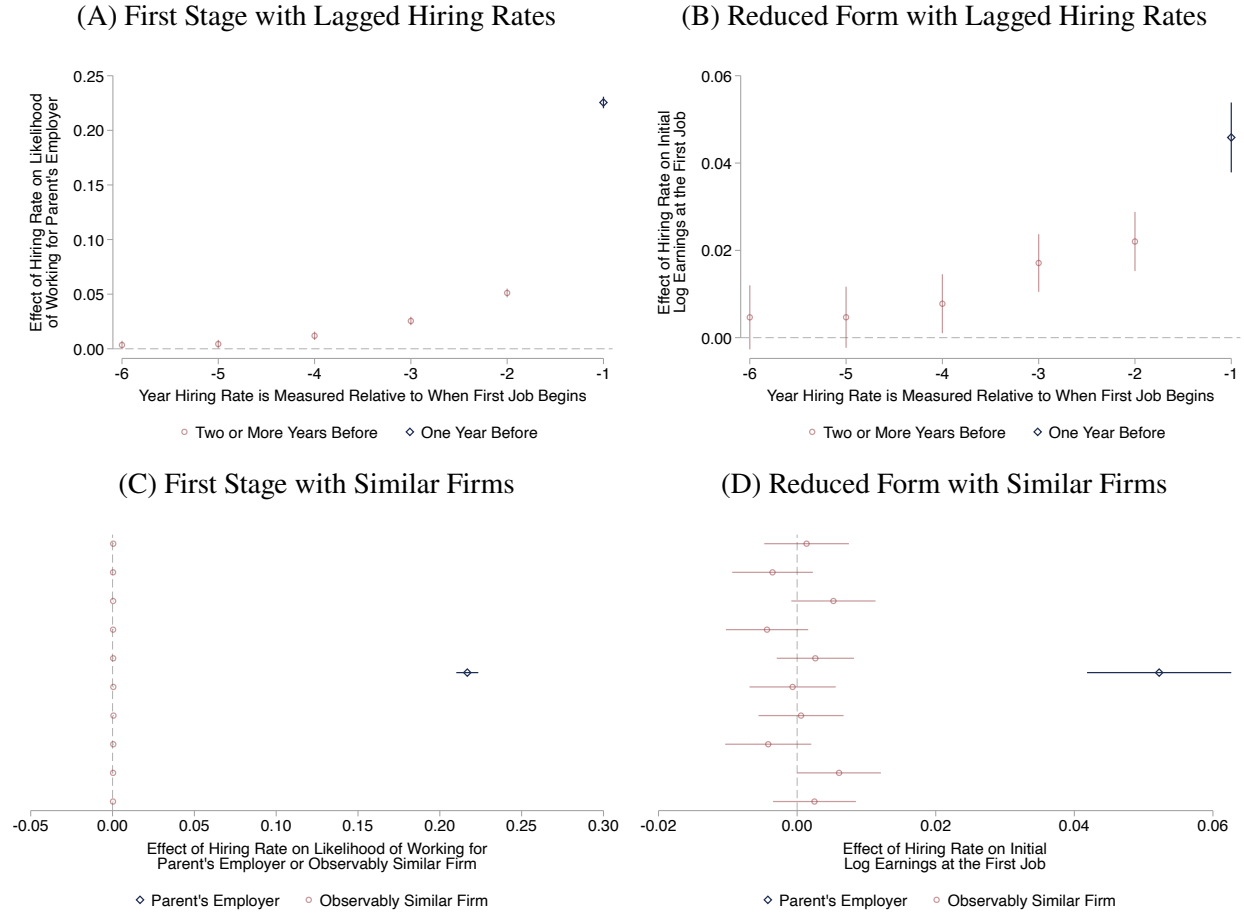
Notes: Column 1 presents the observed IGE, which is the coefficient from a regression of the log initial earnings of the young worker on the log earnings of the parent. Columns 2 through 6 present estimates of the change in the IGE if no one worked for a parent's employer (see Equation 7), which is presented as a percent change relative to the observed IGE. Column 2 presents the baseline estimates. Columns 3 and 4 present estimates assuming that the effect on earnings or the use of connections does not vary by parental earnings, respectively. Column 5 assumes that 20 percent of individuals who work for a parent's employer would have obtained these jobs absent connections, and multiplies the share of individuals who work for a parent's employer by 0.8. Column 6 uses estimates of the earnings gains from the mass layoff design. Standard errors are calculated using the delta method and are presented in parentheses.

Figure 1: Likelihood of Working for Parent's Current vs. Future Employer



Notes: The horizontal axis defines a sample of individuals based on when their parent started working for a new firm. The sample is limited to parents who remain at these new jobs for at least 12 quarters. Thus, the blue diamond markers represent cases in which the parent recently joined and currently works for the firm when their child starts their first stable job. The red circle markers represent cases in which the parent will join the firm in the near future but is not currently working there when their child starts their first stable job. Each point plots the proportion of individuals who work for their parent's current or future employer.

Figure 2: Identifying Variation: Evidence from Lagged Hiring Rates and Similar Firms



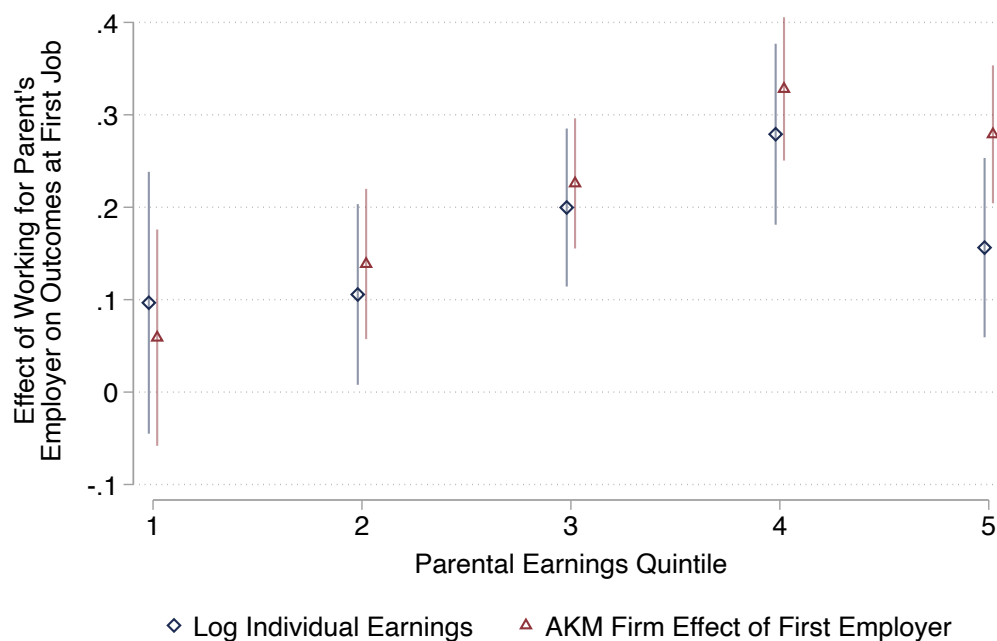
Notes: Panels A and B present estimates using lagged hiring rates at the parent's employer. Each point represents an estimate from a regression of an outcome variable on the hiring rate at the parent's employer. The outcome variable is an indicator for whether the individual works for their parent's employer in Panel A and initial log earnings in Panel B. The horizontal axis defines the time at which the hiring rate at the parent's employer is measured. Panels C and D present estimates using hiring rates at firms that are observably similar to the parent's employer. Each point represents an estimate from a separate regression of an outcome variable on the hiring rate at the parent's employer (blue diamond) or an observably similar firm (red circle). The observably similar firm is matched to the parent's employer by commuting zone, two-digit industry, and size class (above or below 500 employees). The outcome variable is an indicator for whether the individual works for the parent's employer or the observably similar firm in Panel C and initial log earnings in Panel D. All regressions include fixed effects for the parent's employer; fixed effects for the local labor market of the parent's employer, defined as the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. Standard errors are clustered at the level of the parent's employer, and the vertical bars represent 95 percent confidence intervals.

Figure 3: Works for Parent's Employer by Parental Earnings



Notes: The figure plots the proportion of individuals whose first stable job is at the same firm as either parent for each percentile of the parental earnings distribution.

Figure 4: Effect on Initial Outcomes at First Job by Parental Earnings



Notes: Each point represents an estimate of the effect of working for a parent's employer based on a separate regression, which is estimated on one of five distinct samples defined by the quintile of parental earnings. The outcome is either individual log earnings at the first stable job or the AKM firm effect associated with the first employer. The instrumental variables specification, defined in Equation 4, uses the hiring rate at the parent's employer as an instrument and controls for fixed effects for the parent's employer; fixed effects for the local labor market (defined by the interaction between commuting zone, industry, and quarter); and log parental earnings, measured in the year before the child begins their first job. Standard errors are clustered at the level of the parent's employer and the vertical bars denote the 95 percent confidence intervals.

Appendix A Additional Tables and Figures

Table A.1: Sample Restriction Criteria

Exclusion Criteria	Observations Remaining	
	Number	Percent
None (sample frame with no restrictions)	47,556,000	100
Child not assigned a unique PIK	38,701,000	81
Unable to link child to parents because either parent is not assigned a unique PIK or the household contains more than 15 people	35,375,000	74
Combined earnings of the parents does not exceed \$15,000	31,693,000	67
The child does not find a stable job by 2018	25,860,000	54

Notes: This table describes the sample restrictions applied to the sample frame. The first column describes the criteria and the second column presents the rounded number of observations that remain after dropping the observations that meet the criteria. These numbers represent a cumulative count after all sample restrictions described in preceding rows are applied. The third column presents this information as a percent of the total sample frame.

Table A.2: Selection into Working for a Parent's Employer by Educational Attainment

	Works for Parent's Employer	Initial Log Earnings at the First Job	
	(1)	(2)	(3)
High School	-0.009 (0.007)		0.026* (0.013)
Some College	-0.029*** (0.007)		0.010 (0.013)
Bachelor's Degree or Higher	-0.031*** (0.007)		0.136*** (0.013)
Works for Parent's Employer		0.038*** (0.009)	0.041*** (0.009)
R-Squared	0.595	0.646	0.651
Observations	83,000	83,000	83,000

Notes: Each column presents estimates from a separate regression. The outcome is defined by the column header and is either an indicator for working for a parent's employer or initial log earnings. All specifications include fixed effects for the household, the quarter in which the young worker starts their first job, and the interaction between race, sex, and year of birth. The omitted education category is "less than high school." Standard errors are presented in parentheses.

*** $p \leq 0.001$, ** $p \leq 0.01$, * $p \leq 0.1$

Table A.3: Effect of Working for Parent's Employer Using Contemporaneous Hiring Rate

	Works for Parent's Employer		Initial Log Earnings
	(1)	(2)	(3)
Hiring Rate at Parent's Employer	0.26*** (0.002)		
LOO Hiring Rate at Parent's Employer		-0.13*** (0.00)	
Works for Parent's Employer			0.18*** (0.009)
First-Stage F-Statistic			24,330
Observations (Millions)	17.81	17.81	17.81

Notes: Columns 1 and 2 present first-stage estimates using different measures of the hiring rate at the parent's employer as the instrument. In column 1 the instrument is the hiring rate at the parent's employer measured in the quarter in which the young worker enters the labor market. In column 2 the instrument is the leave-one-out (LOO) hiring rate. Column 3 presents estimates from the second-stage using the observed hiring rate at the parent's employer as the instrument. The outcome in column 3 is the log earnings of the young worker. All regressions include fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, where the local labor market is defined by the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. Standard errors are clustered at the level of the parent's employer and are presented in parentheses.

*** $p \leq 0.001$, ** $p \leq 0.01$, * $p \leq 0.1$

Table A.4: Analysis of Displaced Workers

	Pre-Displacement Log Earnings		Change in Log Earnings		Change in AKM Firm Effect
	(1)	(2)	(3)	(4)	(5)
Works for Parent's Employer	-0.129*** (0.012)	-0.037*** (0.010)	0.223*** (0.017)	0.208*** (0.015)	0.188*** (0.005)
Employer Fixed Effects		X	X	X	X
Individual-Level Covariates				X	X
Observations	313,000	313,000	313,000	313,000	308,000

Notes: Each column presents estimates from a separate regression estimated via OLS. The outcome variable is defined by the column header and the main independent variable is an indicator for whether the individual joined their parent's employer in their first job after displacement. Columns 2 through 5 control for a fixed effect for the employer from which the individual was displaced. Columns 4 through 5 additionally control for the average of the outcome variable in the four quarters prior to displacement, parental earnings rank, fixed effects for age, and fixed effects for the interaction between race, sex, and year of birth. The sample includes young workers displaced by a mass layoff. Standard errors are presented in parentheses.

*** $p \leq 0.001$, ** $p \leq 0.01$, * $p \leq 0.1$

Table A.5: Effect of Working for the Parent's Employer Three Years After Entry

	Annual Earnings After Entry		
	One Year (1)	Two Years (2)	Three Years (3)
Works for Parent's Employer	0.24*** (0.02)	0.23*** (0.03)	0.16*** (0.03)
Mean	26,760	26,870	28,100
Standard Deviation	14,790	19,180	21,630
First-Stage F-statistic	5,883	5,883	5,883
Observations (Millions)	14.98	14.98	14.98

Notes: This table reports estimates of the effects on annual earnings one to three years after individuals begin their first job. The outcomes in Columns 1–3 are annual earnings measured one, two, and three years after first-job start, respectively. All specifications include fixed effects for the parent's employer; fixed effects for the local labor market of the parent's employer, defined as the interaction of commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. All regressions are estimated using Poisson models with a control function analogous to the specification in equation 4. Standard errors are clustered at the parent-employer level and are reported in parentheses. The reported means and standard deviations of the outcome are calculated among individuals who do not work for their parent's employer.

*** $p \leq 0.001$, ** $p \leq 0.01$, * $p \leq 0.1$

Table A.6: Effect of Working for Parent's Employer by Parental Earnings

	Parental Earnings Quintile				
	First (1)	Second (2)	Third (3)	Fourth (4)	Fifth (5)
A. Instrumental Variables Estimator					
<i>A.1. Effect on Individual Log Earnings at First Job</i>					
Works for Parent's Employer	0.097 (0.072)	0.106* (0.050)	0.200*** (0.044)	0.279*** (0.050)	0.156** (0.050)
First-Stage F-Statistic	444.5	918.4	1,264	1,118	1,319
Observations in Millions	1.927	2.769	3.386	3.765	3.73
<i>A.2. Effect on AKM Firm Effect of First Employer</i>					
Works for Parent's Employer	0.059 (0.060)	0.139*** (0.041)	0.226*** (0.036)	0.328*** (0.040)	0.279*** (0.038)
First-Stage F-Statistic	436.8	907.3	1,245	1,106	1,318
Observations in Millions	1.914	2.75	3.364	3.74	3.7
B. Mass Layoff Estimator					
<i>B.1. Effect on Change in Individual Log Earnings After Displacement</i>					
Works for Parent's Employer	0.114* (0.053)	0.215*** (0.049)	0.160** (0.049)	0.201*** (0.045)	0.258*** (0.045)
Observations	62,000	65,000	66,000	65,000	55,000
<i>B.2. Effect on Change in AKM Firm Effect After Displacement</i>					
Works for Parent's Employer	0.069*** (0.015)	0.173*** (0.014)	0.171*** (0.015)	0.247*** (0.013)	0.274*** (0.015)
Observations	61,000	64,000	65,000	64,000	54,000

Notes: Each column presents estimates from a separate regression. The sample in Columns 1 through 5 is restricted to individuals whose parents are in the first through fifth quintile of the earnings distribution, respectively. Panel A presents estimates from the instrumental variables estimator defined by equation 4. The outcomes in Panels A.1 and A.2 include individual log earnings at the first stable job and the AKM firm effect associated with the first employer, respectively. The estimates are from the second stage and working for the parent's employer is instrumented for using the hiring rate at the parent's employer. All regressions include fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, where the local labor market is defined by the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. Panel B presents estimates from the mass layoff estimator defined by equation D.1. The outcomes in Panels B.1 and B.2 include the change in individual log earnings after displacement and the change in the AKM firm effect, respectively. The main independent variable is an indicator for whether the individual joined their parent's employer in their first job after displacement. All specifications control for a fixed effect for the employer from which the individual was displaced, the average of the outcome variable in the four quarters prior to displacement, parental earnings rank, fixed effects for age, and fixed effects for the interaction between race, sex, and year of birth. The sample includes young workers displaced by a mass layoff. Standard errors are presented in parentheses.

*** $p \leq 0.001$, ** $p \leq 0.01$, * $p \leq 0.1$

Table A.7: Intergenerational Elasticity of Earnings

	Log Initial Earnings	Log Average Earnings Ages 29-31		
	(1)	(2)	(3)	(4)
Log Parental Earnings	0.136 (0.000)	0.482 (0.002)	0.162 (0.000)	0.491 (0.002)
Sample Excludes Low-Earners	Yes	No	Yes	No
Measure of Parental Earnings	Long-Run	Long-Run	Long-Run	Age 16-20
Observations (Millions)	25.860	7.619	5.150	7.073

Notes: Each column presents estimates from a separate regression of the log earnings of the child on the log earnings of the parent. In column 1 the earnings of the child are measured at the first job. In columns 2-4 the earnings of the child are measured as the average annual earnings between the ages of 29 and 31. In columns 1-3 parental earnings corresponds to the long-run measure described in the text. In column 4 parental earnings corresponds to the average earnings of the parents in the years when their child was between the ages of 16 and 20 and the sample excludes observations if the combined earnings of the parents is less than \$15,000. Columns 1 and 3 exclude children with sufficiently low earnings, while columns 2 and 4 add one to earnings in order to retain zeros.

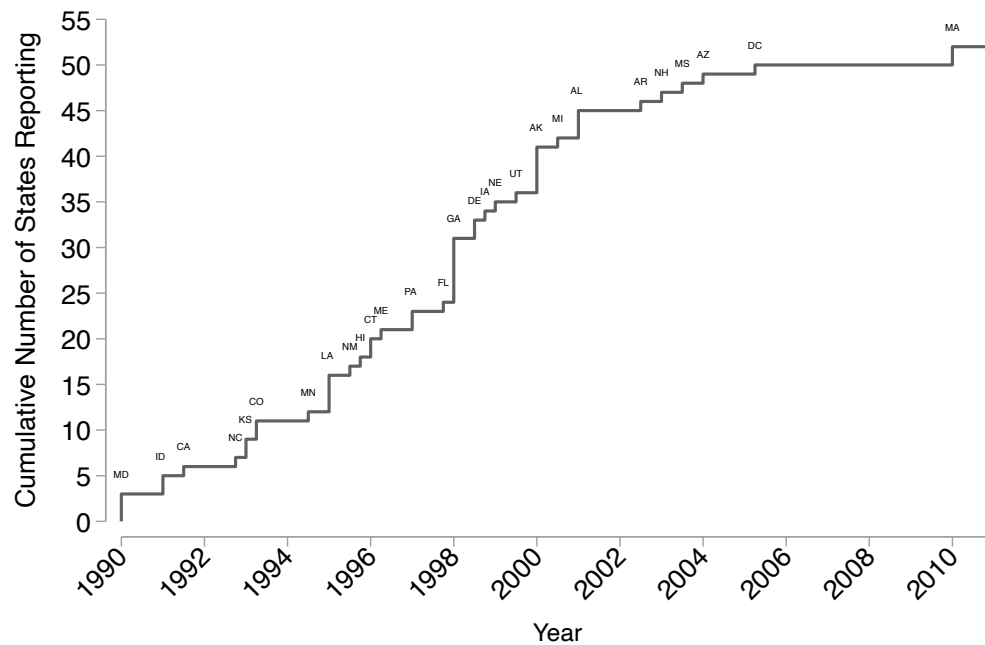
Table A.8: Effect of Working for Parent's Employer by Sex of Parent and Child

	Initial Log Earnings			
	(1)	(2)	(3)	(4)
Works for Parent's Employer	0.28*** (0.07)	0.23*** (0.03)	0.18*** (0.04)	0.30*** (0.05)
Sex of Parent	Father	Father	Mother	Mother
Sex of Child	Daughter	Son	Daughter	Son
Proportion Who Work for Parent's Employer	0.015	0.039	0.034	0.026
First-Stage F-Statistic	1,178	2,946	1,465	1,188
Observations (Millions)	4.96	5.22	5.69	5.73

Notes: This table presents second-stage estimates from the two-stage least squares specification defined in equation 4. Each column presents estimates based on a distinct sample defined by the sex of the parent and the sex of the child. The outcome is the log earnings of the young worker and all regressions include fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, where the local labor market is defined by the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. Standard errors are clustered at the level of the parent's employer and are presented in parentheses.

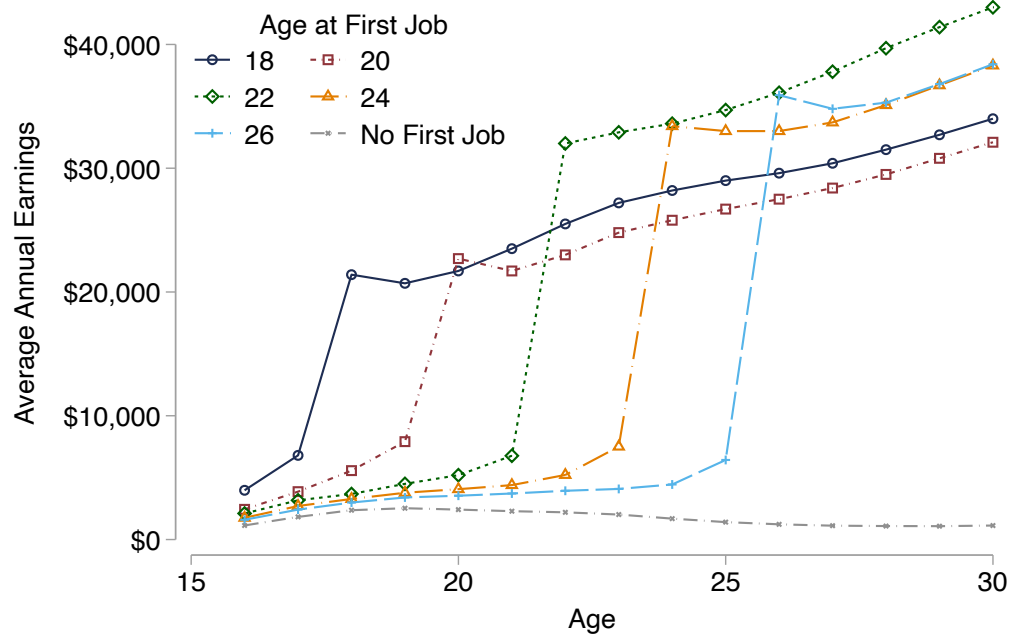
*** $p \leq 0.001$, ** $p \leq 0.01$, * $p \leq 0.1$

Figure A.1: States Covered by the LEHD Data



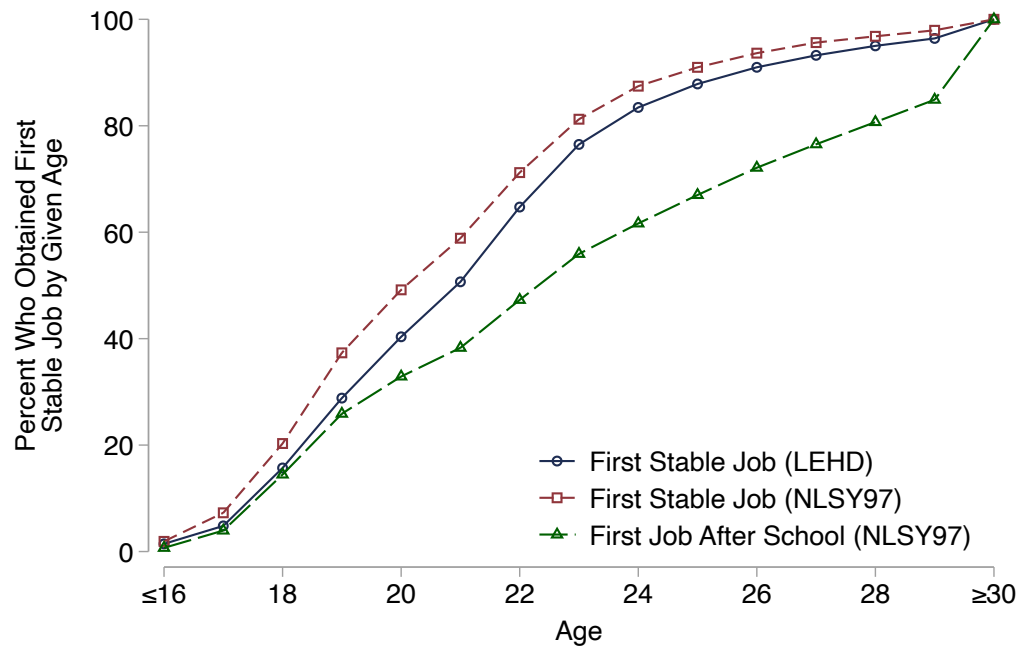
Notes: The figure plots the number of states that are reporting to the Longitudinal Employer-Household Dynamics (LEHD) program by the year listed on the horizontal axis. The abbreviations above the solid line represent the states that begin reporting in that year.

Figure A.2: Age-Earnings Profile



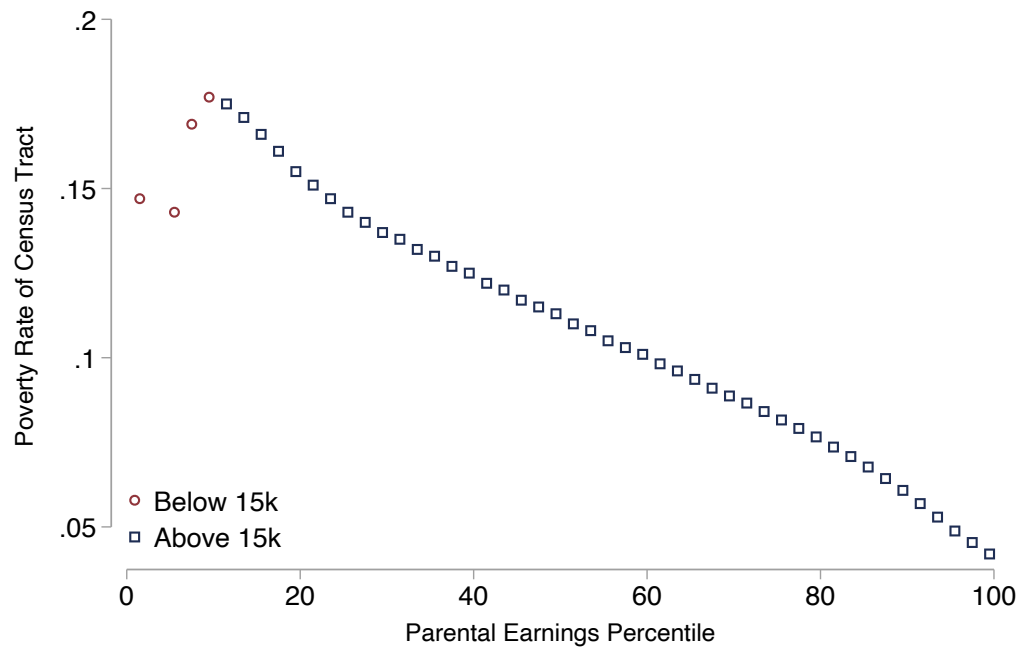
Notes: The figure plots the average annual earnings by age for different groups of workers defined by the age they were when they found their first stable job. The sample includes individuals who turned 30 by 2018.

Figure A.3: Age of Entry



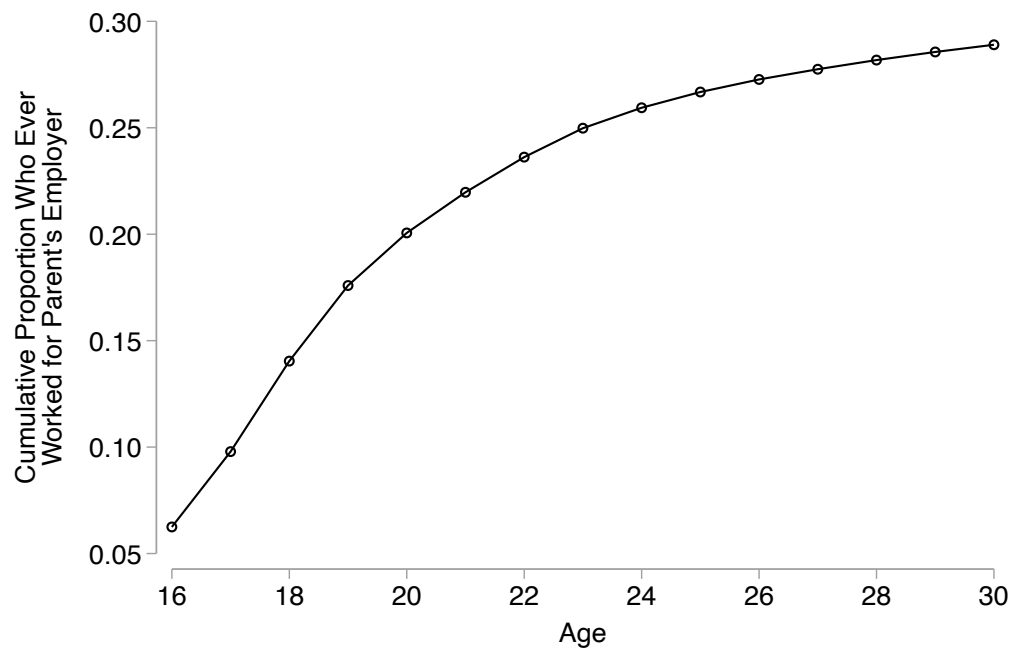
Notes: The figure plots the cumulative proportion of children that have entered the labor market by the age indicated on horizontal axis. For comparison, I also plot results using alternative measures of entry constructed from the NLSY97. These measures include the first stable job (working at least 35 hours for 36 consecutive weeks) and the first stable job after all schooling is completed.

Figure A.4: Parental Earnings and Neighborhood Poverty



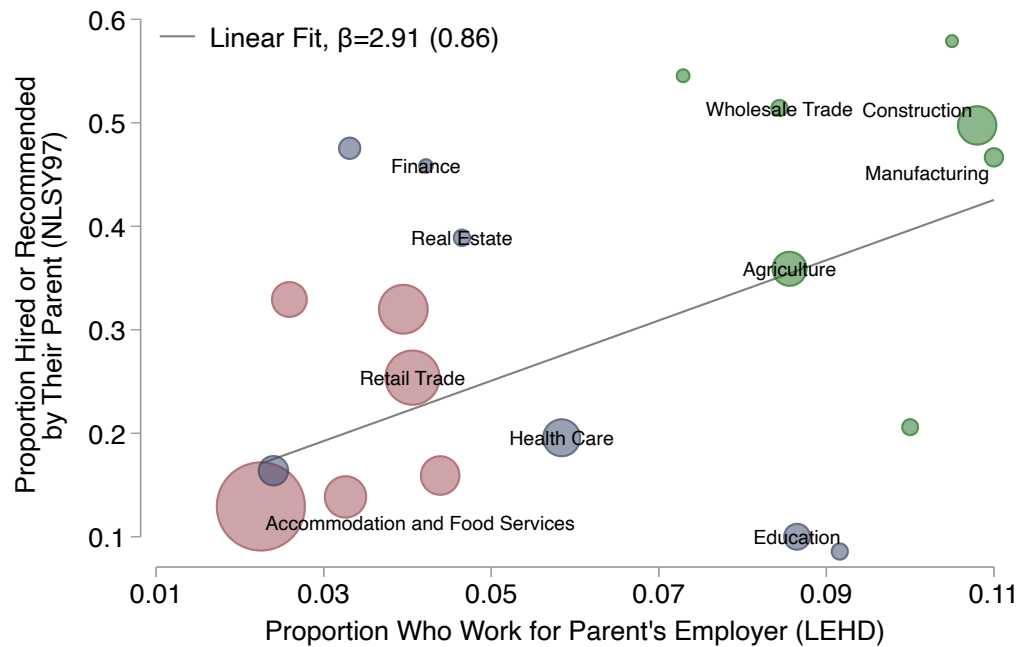
Notes: The figure plots the average poverty rate of the Census tract in which the parents lived in 2000. Parents are grouped into 50 equal-sized bins based on their earnings and each point represents a statistic for one of these distinct samples.

Figure A.5: Likelihood of Working for Parent's Employer by Age 30



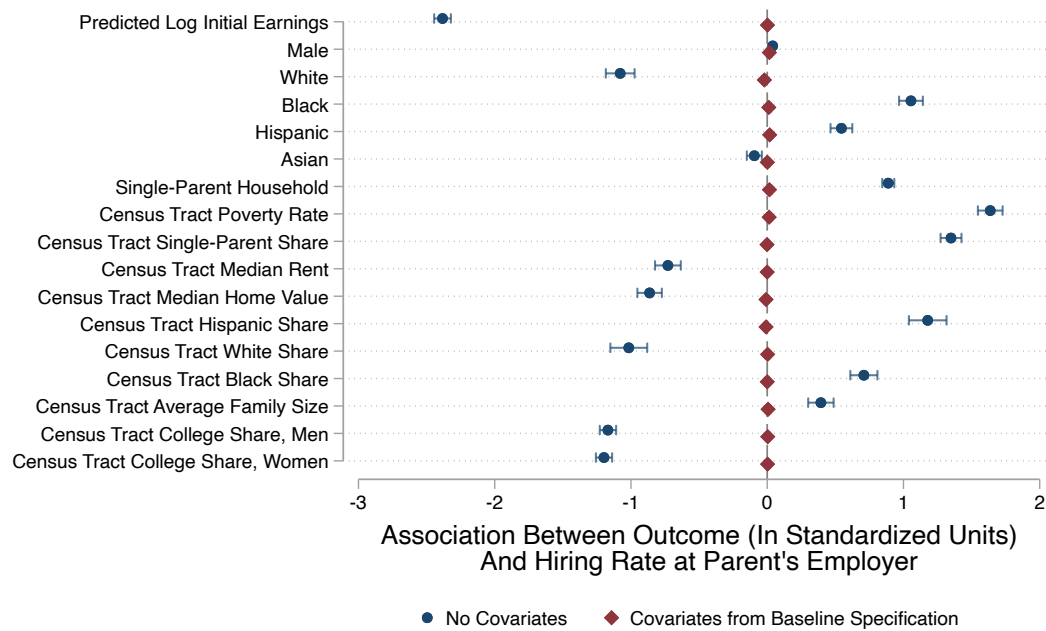
Notes: This figure presents the cumulative share of workers who ever worked for a parent's employer between the ages of 16 and 30. The statistics are calculated on the sample of individuals who turn 30 by 2018.

Figure A.6: Industry-Level Association with Use of Social Contacts



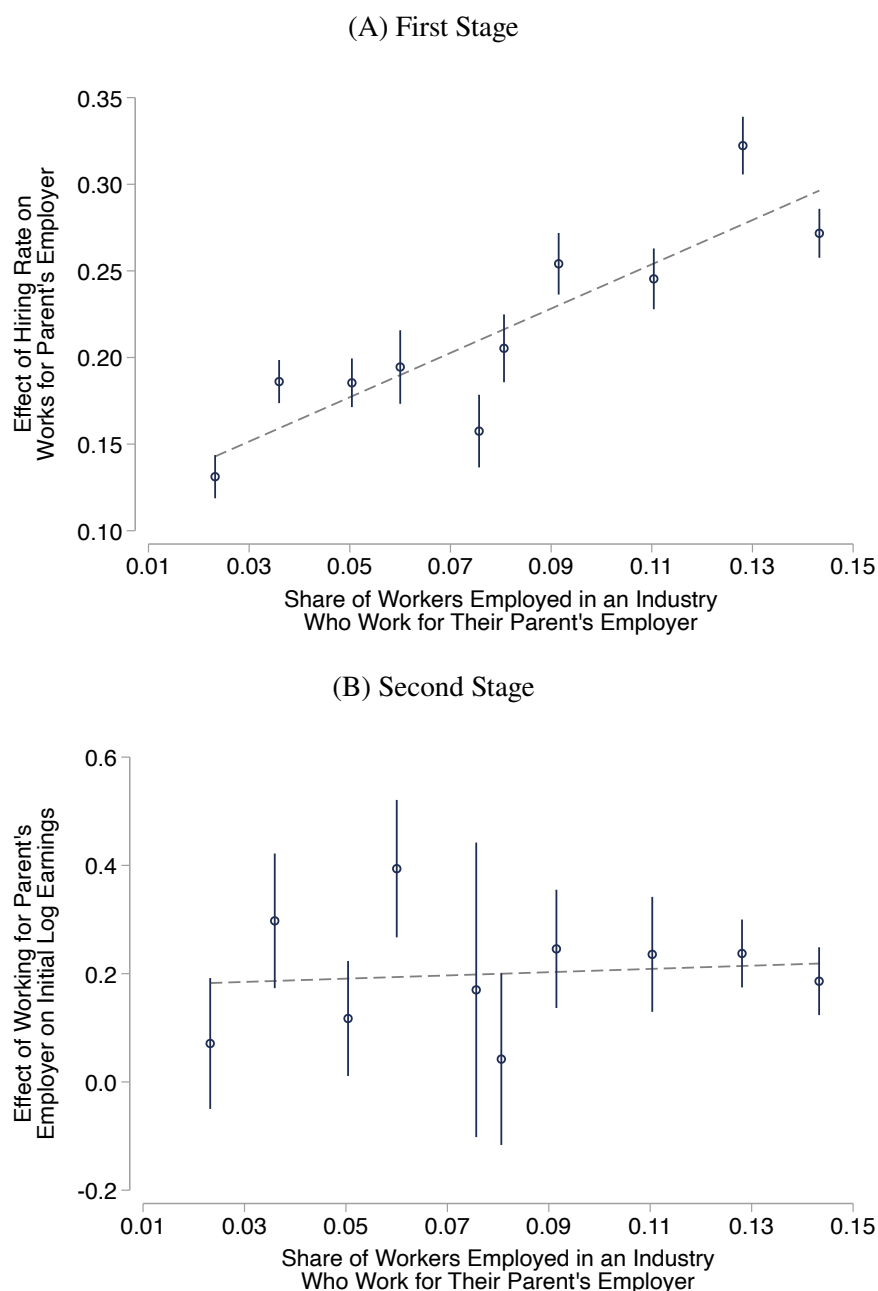
Notes: Each point represents an industry-level statistic, with marker size proportional to sample size. The horizontal axis shows the share of individuals whose first job was at a parent's employer, measured using the LEHD and 2000 Decennial Census. The vertical axis shows the share of jobs in which individuals were hired or recommended by a parent, based on data from the NLSY97. The linear fit is depicted by the gray line and the coefficient and standard error appear in the legend.

Figure A.7: Association Between Hiring Rate and Baseline Characteristics



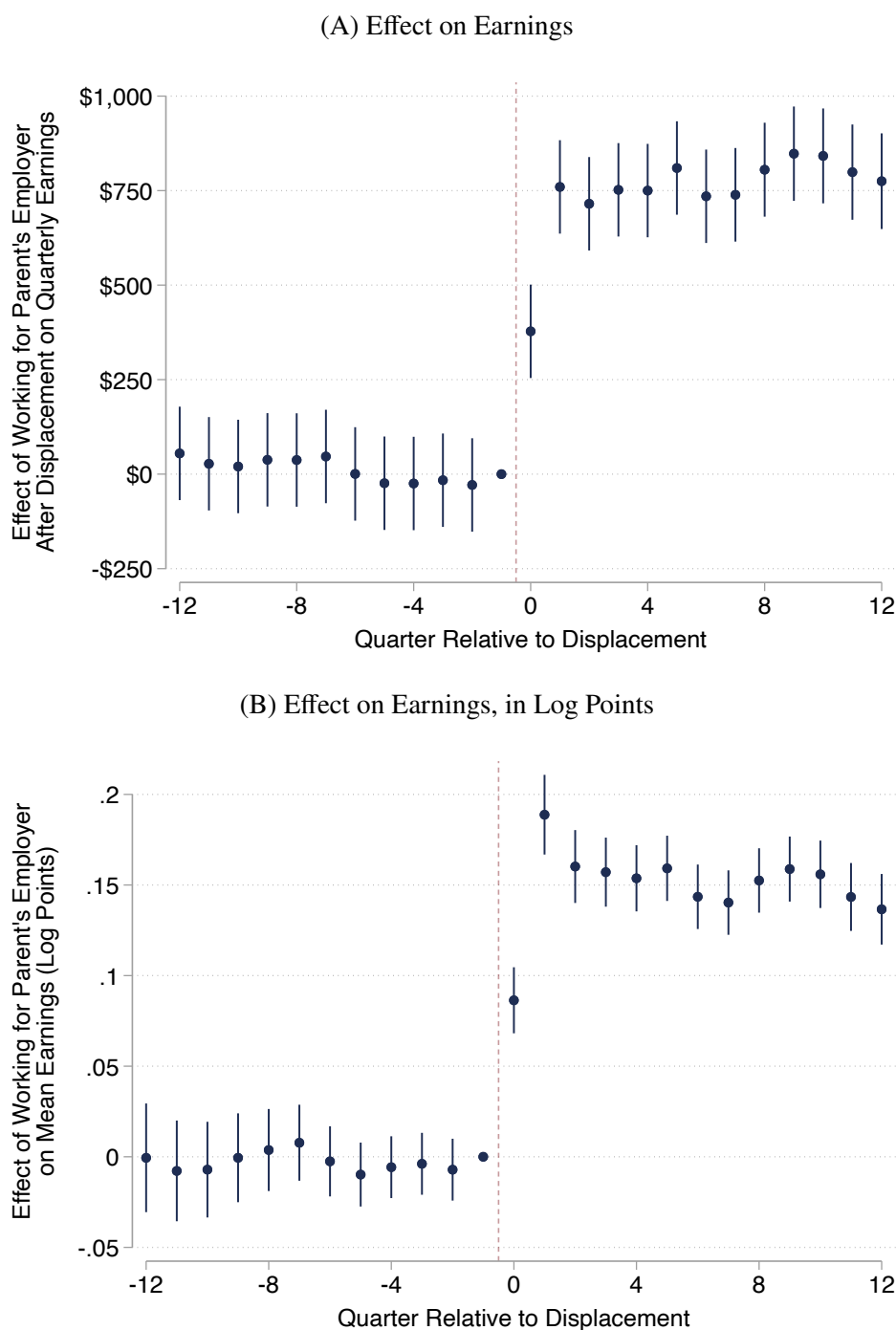
Notes: This figure plots coefficients from regressions of each baseline characteristic on the hiring rate at the parent's employer. The blue circles show estimates from specifications with no controls; the red diamonds show estimates from the baseline specification in Equation 4 with the following covariates: fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, where the local labor market is defined by the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. Census tract characteristics correspond to the tract in which the child lived in 2000. The outcome in the first row is predicted initial earnings, constructed using the baseline characteristics shown in the remaining rows. Bars denote 95% confidence intervals.

Figure A.8: Heterogeneity by Industry-Level Measures of the Use of Parental Connections



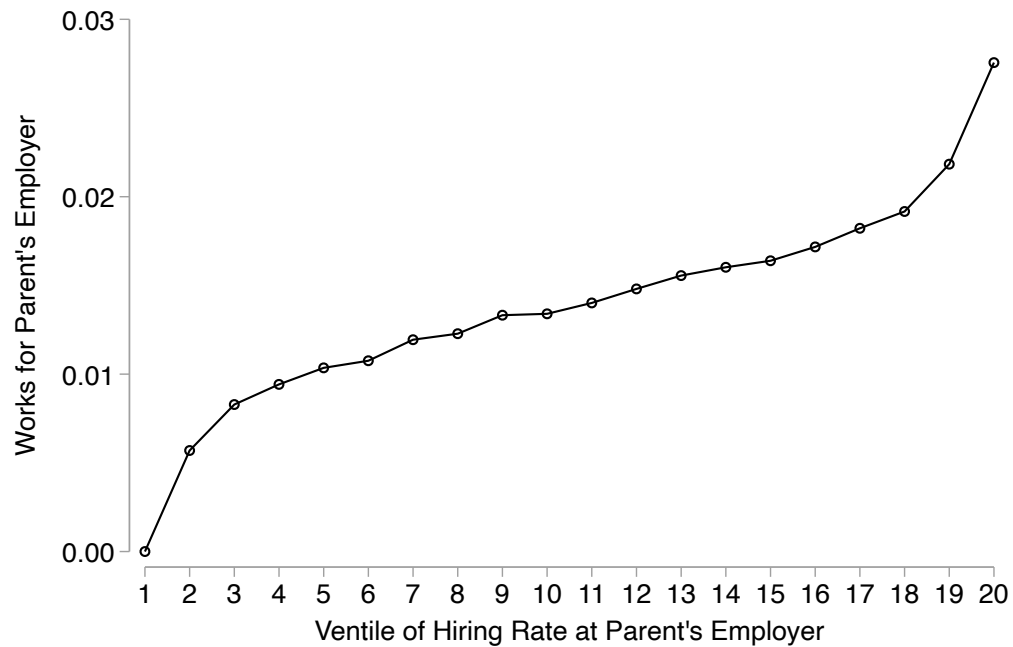
Notes: Industries are grouped into deciles based on the share of young workers in that industry who work for a parent's employer. Panel A presents first-stage estimates and regresses an indicator for whether the individual works for their parent's employer on the interaction between the hiring rate at the parent's employer and the industry decile of the parent's employer. The estimated coefficients are plotted against the share of workers in that industry decile who work for their parent's employer. Panel B presents the analogous second-stage estimates, where the outcome variable is initial log earnings. Both regressions include fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, where the local labor market is defined by the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. Standard errors are clustered at the level of the parent's employer and the vertical bars represent 95 percent confidence intervals.

Figure A.9: Analysis of Displaced Workers, Event-Study Specification



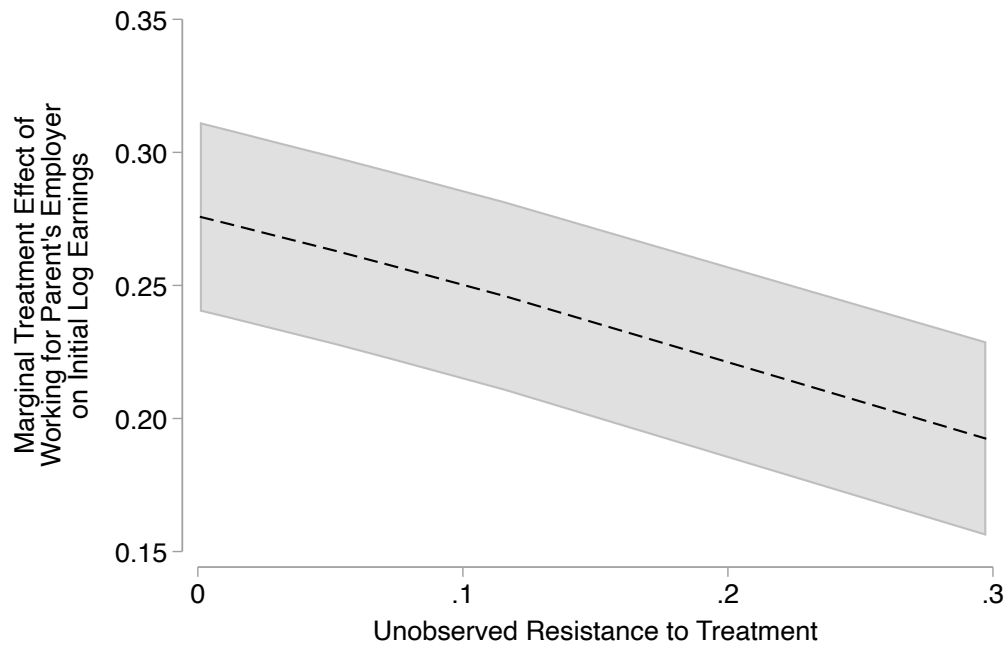
Notes: The figures represent estimates from the event-study specification described in Equation D.2, which regresses individual log earnings on the interaction between quarter relative to displacement and an indicator for whether the individual finds a job at the parent's employer after displacement. The specification controls for a quadratic in age and fixed effects for the employer prior to displacement. The sample includes young workers who were displaced by a mass layoff and is a balanced panel containing data 12 quarters before and after the mass layoff event. Panel A is estimated via OLS and Panel B is estimated using a Poisson model. Standard errors are clustered at the level of the mass layoff event and the vertical bars denote the 95 percent confidence interval.

Figure A.10: Visualization of First Stage



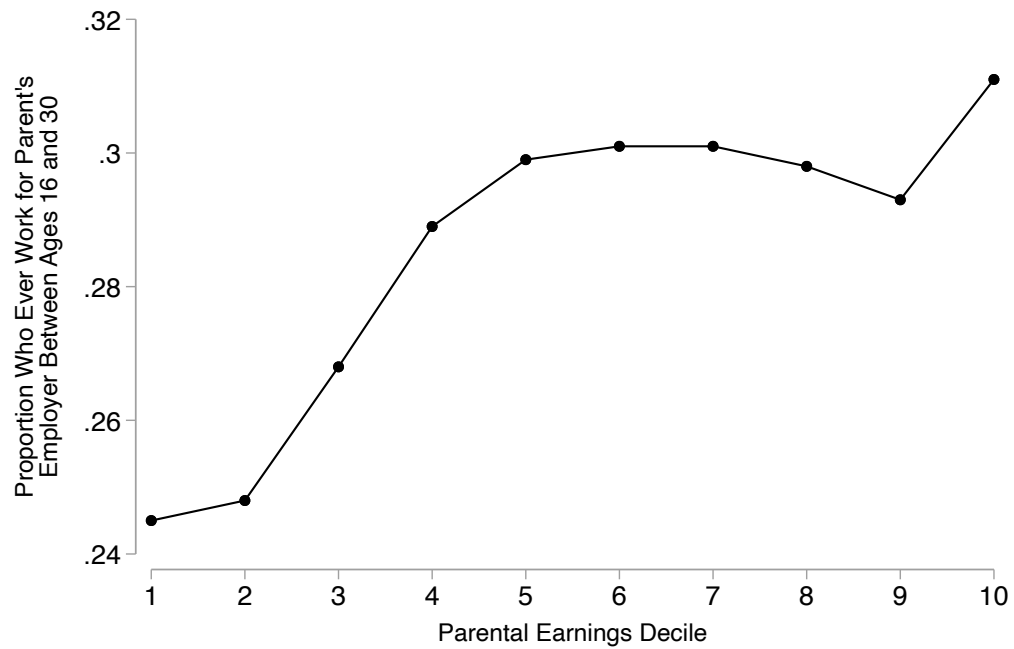
Notes: This figure presents a graphical visualization of the first stage. The hiring rate at the parent's employer and the indicator for working for the parent's employer are regressed against the standard set of covariates in Equation 4 including fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, where the local labor market is defined by the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. The data are grouped into ventiles based on the residuals from this regression. The figure presents the average value of the residualized indicator for working for the parent's employer for each ventile of the residualized hiring rate.

Figure A.11: Marginal Treatment Effects Curve



Notes: This figure plots estimates of the marginal treatment effect of working for a parent's employer on initial log earnings at the first stable job. I first estimate the propensity score using the first-stage specification in Equation 4, where the hiring rate at the parent's employer is used to predict whether the individual works for the parent's employer. I then regress initial log earnings on a third-order polynomial in the estimated propensity score, controlling for the baseline covariates in Equation 4: fixed effects for the parent's employer; fixed effects for the local labor market in which the parent's employer is located, defined by the interaction between commuting zone, industry, and quarter; and log parental earnings, measured in the year before the child begins their first job. The specification also includes interactions between the estimated propensity score and log parental earnings, the AKM firm pay premium of the parent's employer, and the commuting-zone-level unemployment rate. The marginal treatment effect is calculated by differentiating this fitted relationship with respect to the estimated propensity score and evaluating the interacted covariates at their sample means. The horizontal axis—which indexes unobserved resistance to treatment—is restricted to the range of values between the 5th and 95th percentiles of the estimated propensity score.

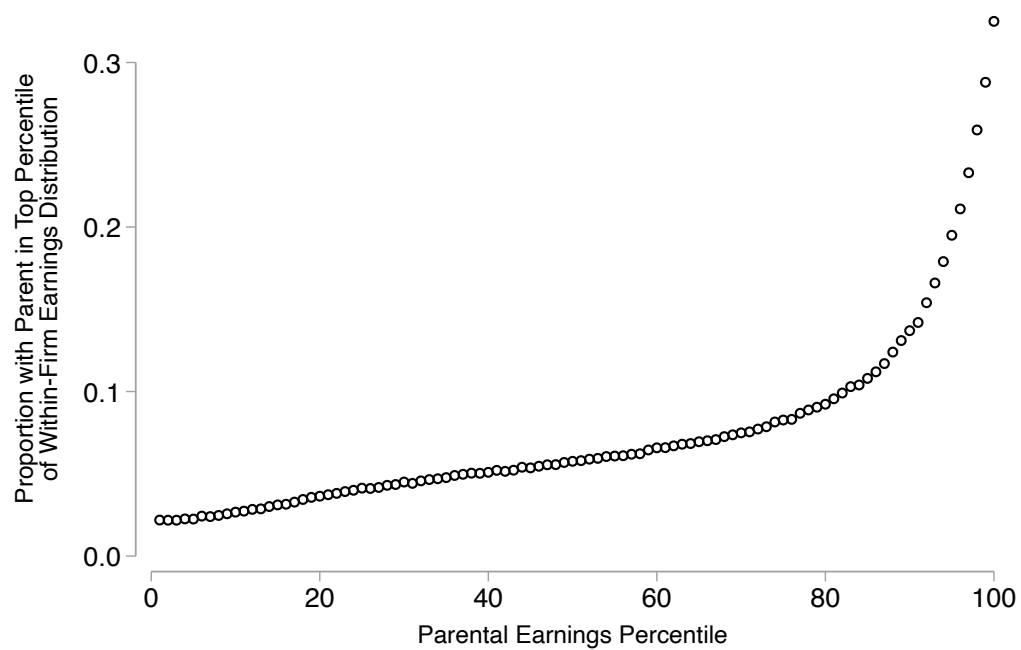
Figure A.12: Works for Parent's Employer Between Ages 16 and 30



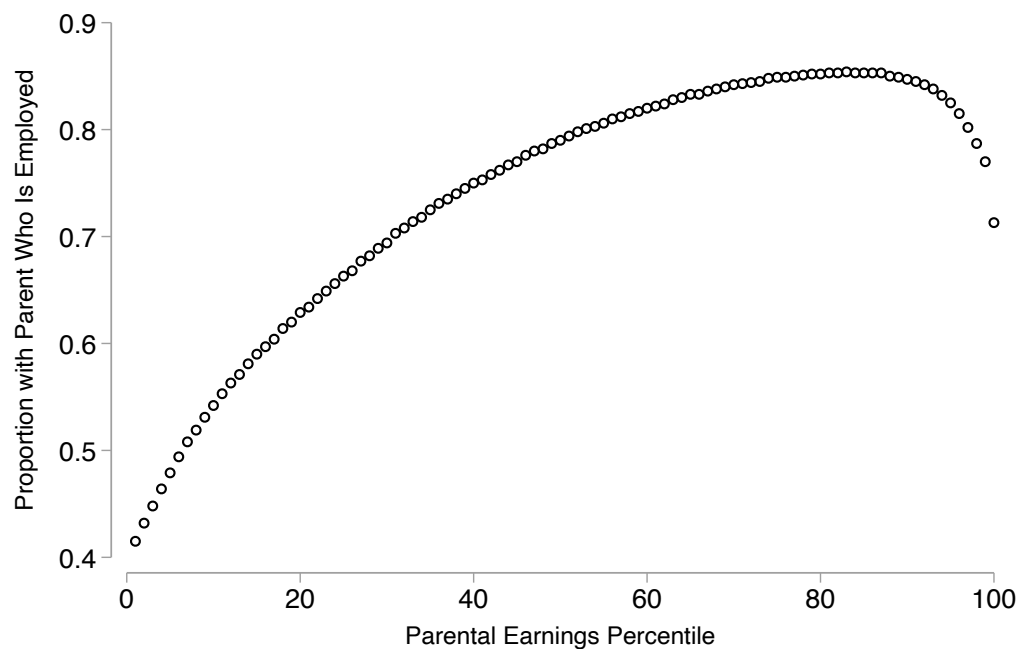
Notes: The figure plots the proportion of individuals who ever work for their parent's employer between the ages of 16 and 30 for each decile of the parental earnings distribution.

Figure A.13: Parent Characteristics by Parental Earnings

(A) Parent is Top Earner at Firm

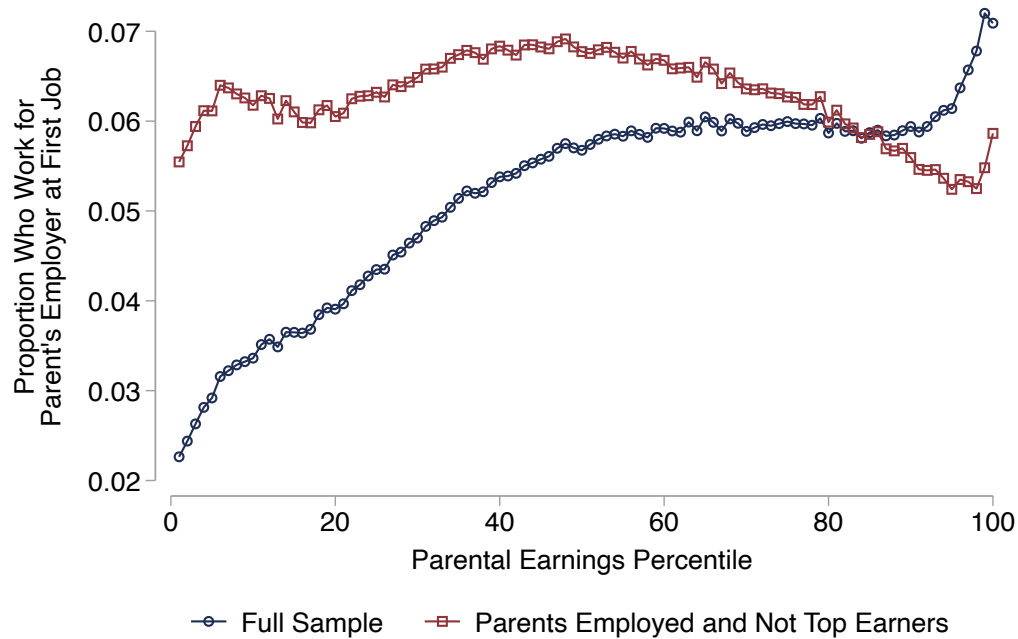


(B) Parent is Employed



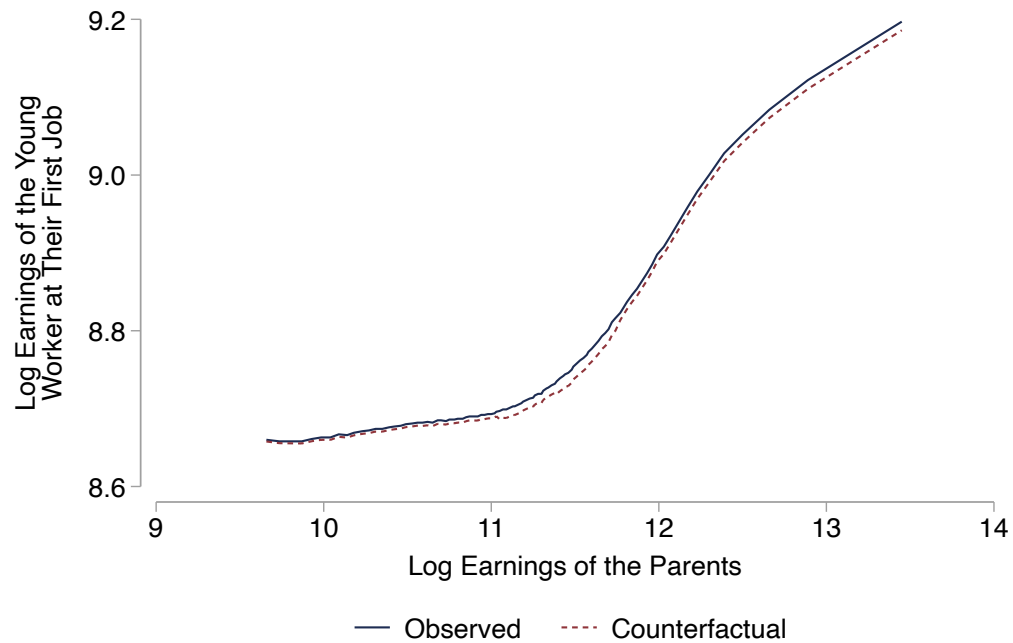
Notes: Panel A plots the proportion of individuals with a parent in the top percentile of the within-firm earnings distribution for each percentile of parental earnings. Panel B plots the proportion of individuals with a parent that is employed for each percentile of parental earnings.

Figure A.14: Works for Parent's Employer by Parental Earnings



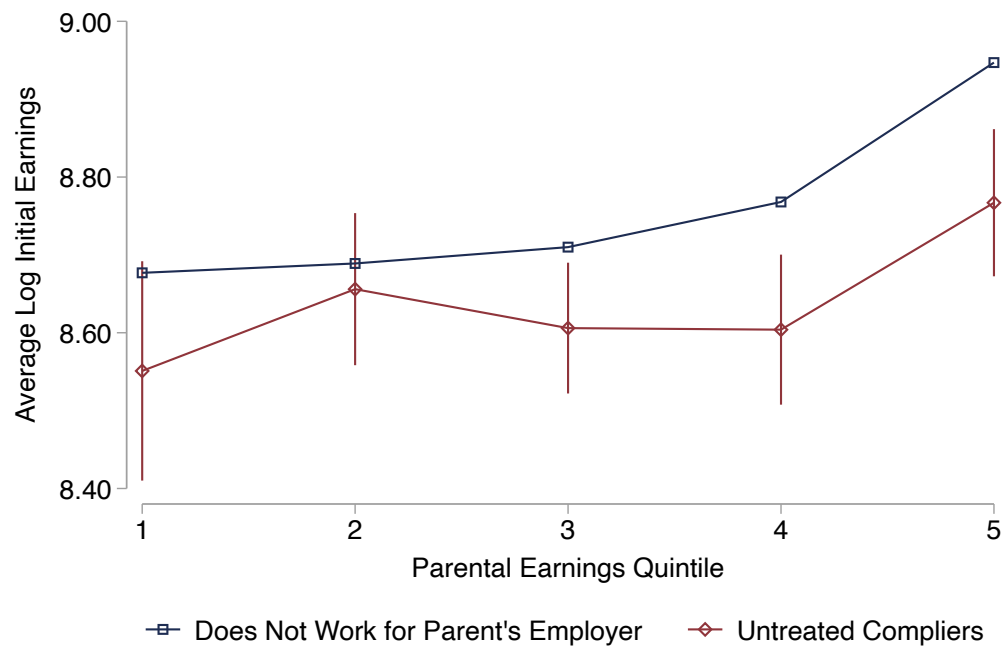
Notes: The figure plots the proportion of individuals whose first stable job is at the same firm as either parent for each percentile of the parental earnings distribution. The navy circles present estimates for the full sample. The red squares present estimates for the sample of individuals who have an employed parent who is not in the top percentile of the within-firm earnings distribution.

Figure A.15: Association Between Log Earnings of Children and Parents



Notes: The figure plots the average initial log earnings of the child against the average log earnings of their parent for each percentile of the parental earnings distribution. The blue solid line represents the observed earnings of the child. The red dashed line represents the counterfactual earnings of the child if no one were to work for a parent's employer.

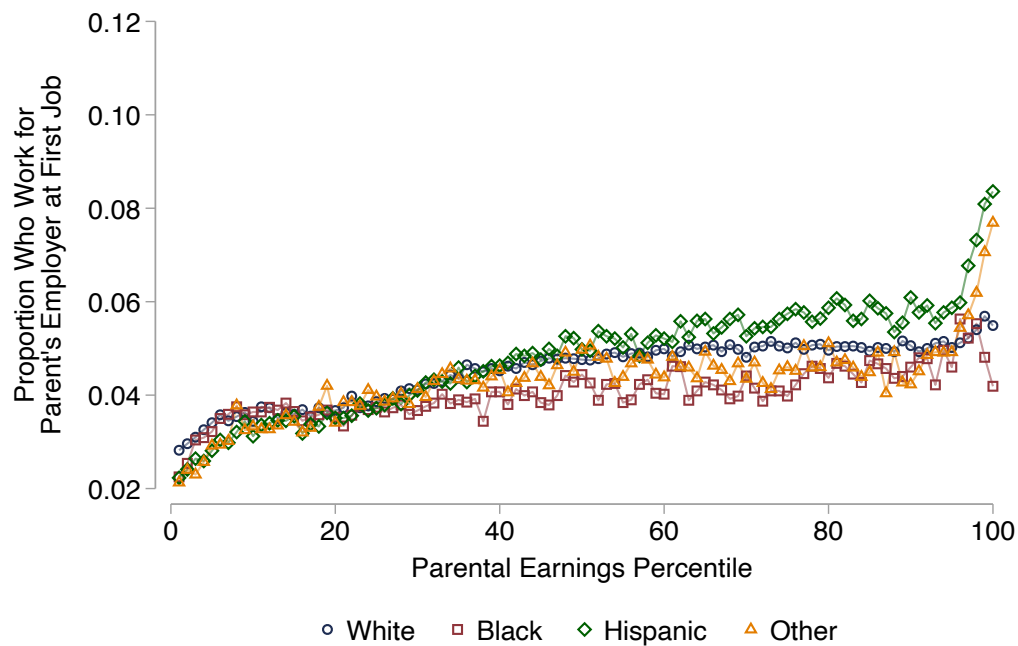
Figure A.16: Mean Earnings for Control and Untreated Compliers by Parental Earnings



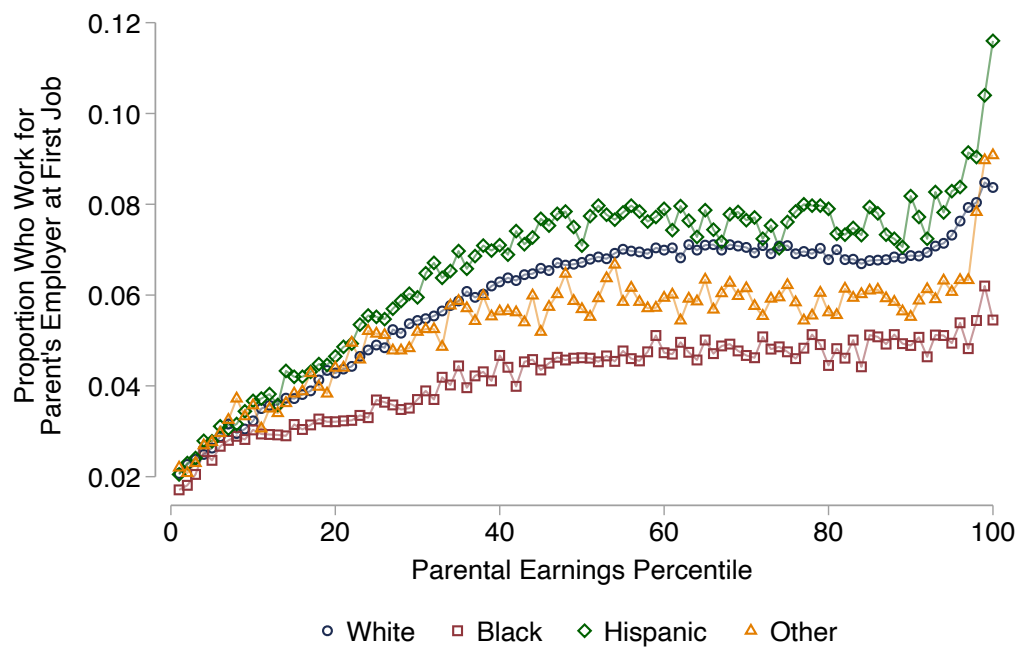
Notes: This figure plots the mean of log initial earnings by parental earnings quintile for those who do not work for their parent's employer and the untreated compliers. The mean of the untreated complier groups is estimated using a variant of Equation 4, where D_i is replaced with $(1 - D_i)$ and the outcome in the second stage is replaced with $y_i \times (1 - D_i)$. The vertical bars denote the 95 percent confidence intervals.

Figure A.17: Works for Parent's Employer by Parental Earnings, Sex, Race, and Ethnicity

(A) Daughters

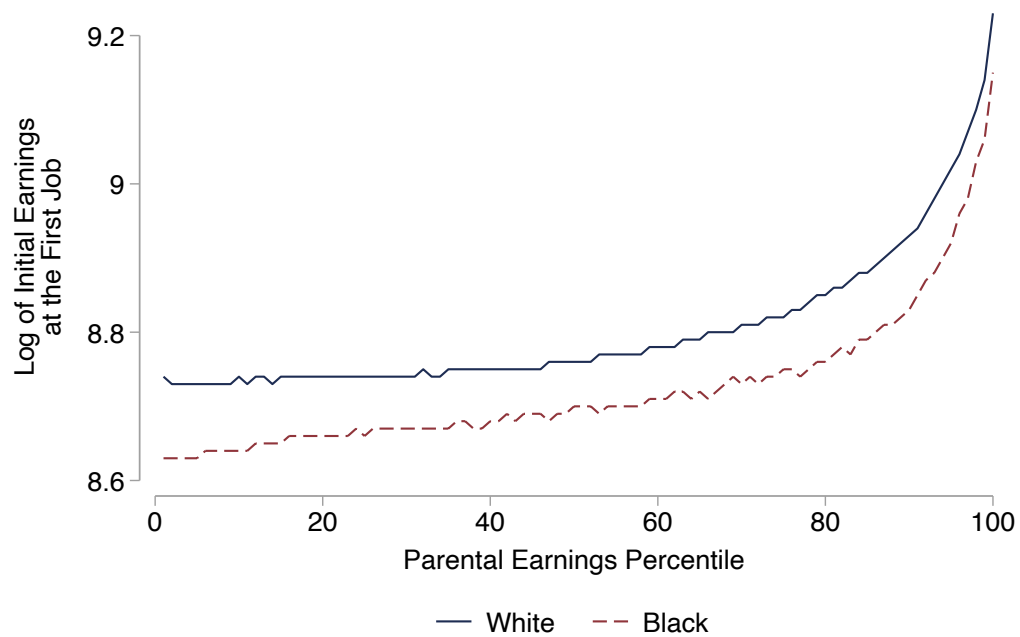


(B) Sons



Notes: Each point represents the proportion of individuals who work for their parent's employer for a sample defined by the interaction between sex, race, ethnicity, and the percentile of the parental earnings distribution.

Figure A.18: Black-White Earnings Gap for Sons



Notes: The figure plots the average initial log earnings of the child for each percentile of parental earnings. The blue solid line and the red dashed line represents the earnings of White and Black sons, respectively.

Appendix B Description of Data

B.1 Measuring Parental Earnings

To estimate the long-run earnings of the parents, I construct a quarterly panel that includes all quarters with strictly positive earnings records between 1990 and 2018 for the parents in the sample. Quarters with zero earnings are not included in the sample. I further restrict the panel to individuals between the ages of 25 and 65 and drop individuals who have fewer than 12 quarters of strictly positive earnings over the entire time period. Parents not included in this panel are assumed to have zero earnings.

Using this quarterly panel, I estimate the following regression:

$$y_{it} = \alpha_i + \beta^g X_{it} + u_{it} \quad (\text{B.1})$$

where i is the individual, t is the quarter, y is total quarterly earnings, α is an individual fixed effect and X is a vector that consists of a third-order polynomial in age. To allow for a flexible age-earnings profile, I estimate this specification separately for groups, g , defined by the interaction between sex, education (less than high school, high school, some college, bachelor's degree or advanced degree), and state of residence in 2000.³⁰

I use the estimates from this model to construct a measure of long-run earnings for each parent. I predict the value of earnings for each quarter and define long-run earnings as the average annual earnings between the ages of 35 and 55. Individuals with either missing or negative values are assigned a long-run earnings of zero. For single-headed households parental earnings is the earnings of the parent. For two-parent households, parental earnings is the sum of the earnings of both parents.

I validate my measure of parental earnings by showing that it strongly correlates with neighborhood poverty rates. Using the Decennial Census, I identify the neighborhood, or Census tract, in which each household lives. Figure A.4 plots the average poverty rate of the neighborhood of residence against the percentile rank of parental earnings. For households with income above \$15,000, there is a negative association between earnings and neighborhood poverty rates. However, this strong relationship breaks down for parents with earnings below \$15,000, which is why I drop these individuals from the sample.

B.2 Grouping Industries into Sectors

I group two-digit North American Industry Classification System (NAICS) industry codes into three distinct sectors, which are defined below. The unskilled service sector includes: retail trade (44,45); administrative and support and waste management and remediation services (56); arts, entertainment and recreation (71); accommodation and food services (72); and other services (81). The skilled service sector includes: information (51); finance and insurance (52); real estate and rental and leasing (53); professional, scientific, and technical services (54); management of companies and enterprises (55); educational services (61); health care and social assistance (62); and public administration (92). The blue-collar sector includes: agriculture, forestry, fishing and hunting (11); mining, quarrying, and oil and gas extraction (21); utilities (22); construction (23); manufacturing (31,32,33); wholesale trade (42); and transportation and warehousing (48,49).

³⁰The education data are either measured using the 2000 Decennial Census long-form and the American Community Surveys, or are imputed (based on earnings) for workers that do not respond to these surveys.

B.3 Firm and Industry Pay Premiums

I estimate firm-specific earnings premiums using the AKM model (Abowd et al., 1999). I estimate the following specification,

$$y_{it} = \alpha_i + \Psi_{j(i,t)} + X_{it}\beta + \epsilon_{it} \quad (\text{B.2})$$

where i is the individual; t is the quarter; y is the log of average quarterly earnings; X_{it} is a vector of time varying controls that include a fixed effect for the quarter and a third order polynomial in age interacted with sex and education; α_i is an individual fixed effect; $\Psi_{j(i,t)}$ is a fixed effect for the employer of i in time t ; and ϵ_{it} is a regression residual.³¹ $\hat{\Psi}_{j(i,t)}$ is the firm pay premium.

I estimate Equation B.2 using a national sample of quarterly earnings records from the LEHD measured between the years 2000 and 2016. The sample includes full quarter jobs for workers between the ages of 15 and 65.³² I drop children from my intergenerational sample. As is standard in the literature, I restrict the sample to the largest connected set. I estimate the model by implementing the iterative method proposed by Guimaraes and Portugal (2010). I am unable to compute the firm pay premium for firms that lie outside of the largest connected set.

I estimate the industry-level premium using similar data and methodology. Because all industries are connected through worker mobility, I estimate the industry premiums on a 10 percent sample of workers and collapse the quarterly data to an annual frequency. In the empirical model I replace the employer fixed effect with a fixed effect for the industry code. I am able to estimate an industry-level pay premium for all industries, and thus there are no missing data for this variable.

References

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- Guimaraes, Paulo, and Pedro Portugal (2010) “A simple feasible procedure to fit models with high-dimensional fixed effects.” *The Stata Journal*, 10(4), pp 628-649.

³¹Identification of the age and time effects in the presence of individual fixed effects is achieved by following Card et al. (2013) and omitting the linear age term for each sex by education group and using a cubic polynomial in age minus 40. This normalization assumes that the age-earnings profile is flat at age 40. While the normalization affects the estimates of the individual fixed effects and the covariate index $X_{it}\beta$, the employer fixed effects are invariant to the normalization used. Data on education comes from the individual characteristics file and is sourced from various surveys and is imputed for many observations.

³²If the worker has multiple jobs in a quarter, I retain the highest-paying job. To limit the influence of outliers, I drop observations if the quarterly earnings exceed one million dollars.

Appendix C Leave-One-Out Hiring Rate

This section explains why the preferred specification uses the lagged hiring rate and not a leave-one-out hiring rate.

Let H_{jt} denote the number of workers that firm j hires in period t and let E_{jt} denote the number of workers employed at the firm. The observed hiring rate is defined as,³³

$$h_{jt}^{obs} = \frac{H_{jt}}{E_{jt}}. \quad (C.1)$$

The concern is that the hiring rate could be mechanically related to an individual being hired by their parent's employer. For individual i whose parent is employed at firm j , let H_{jt}^d denote the number of hires at firm j in period t under treatment status $d \in \{0, 1\}$. Then,

$$H_{jt} = H_{jt}^0 + D_i \Delta, \quad (C.2)$$

where $\Delta = (H_{jt}^1 - H_{jt}^0)$. If $\Delta = 1$, then the firm creates an additional job opening in order to hire the child of a current employee. If $\Delta = 0$, then the firm does not create an additional job opening but hires the child of the current employee instead of a different job seeker. For simplicity, assume that the effect of hiring the child on end-of-period employment is the same as the effect on hires, so that $E_{jt} = E_{jt}^0 + D_i \Delta$, where E_{jt}^0 denotes the firm size absent the child joining the firm.

The ideal hiring rate to use as an instrument would not be a function of whether the child joins that firm and is denoted,

$$h_{jt}^{ideal} = \frac{H_{jt}^0}{E_{jt}^0}. \quad (C.3)$$

This ideal hiring rate is not observable in the data, but an intuitive way to approximate h_{jt}^{ideal} would be to use a leave-one-out hiring rate defined as

$$h_{jt}^{loo} = \begin{cases} h_{jt}^{obs} & \text{if } D_i = 0 \\ \frac{H_{jt}^0}{E_{jt}^0} & \text{if } D_i = 1 \end{cases}. \quad (C.4)$$

The leave-one-out approach is widely used in other contexts. However, it is not a valid approach in the current context because the appropriate adjustment to the observed hiring rate depends critically on whether hiring the child increases total hires and employment, or instead replaces another hire. Under the assumption that $H_{jt}^0 \leq E_{jt}^0$, the following two statements are true:

1. If $\Delta = 1$, then $h_{jt}^{loo} = h_{jt}^{ideal} \leq h_{jt}^{obs}$.
2. If $\Delta = 0$, then $h_{jt}^{loo} \leq h_{jt}^{ideal} = h_{jt}^{obs}$.

In other words, if firms create new jobs to hire the children of current employees, then the leave-one-out measure is equal to the ideal hiring rate. However, if firms do not create new job openings, then the observed hiring rate is equal to the ideal hiring rate. In the latter case, the

³³My empirical analysis uses the End-of-Quarter Hiring Rate definition from the Quarterly Workforce Indicators. The theoretical framework in this section abstracts from this detail to simplify exposition.

leave-one-out hiring rate will mechanically be lower than the ideal hiring rate when the child joins that firm.

The data appear to be more consistent with $\Delta = 0$. Table A.3 presents first- and second-stage estimates using the contemporaneous hiring rate as an instrument—i.e., the hiring rate measure in the same quarter that the young worker finds their first job. Column 1 shows that using the contemporaneous hiring rate yields a strong positive first stage. Column 2 shows that using the leave-one-out hiring rate yields a negative first-stage coefficient, which is the result of the mechanical bias discussed above. Column 3 shows that using the contemporaneous hiring rate as the instrument yields estimated earnings gains that are similar to the preferred specification: 18 log points compared to 21 log points. While the results are consistent with $\Delta = 0$, my preferred approach avoids these issues by using the lagged hiring rate.

Appendix D Analysis of Displaced Workers

This section presents estimates from an alternative empirical strategy, which compares young workers who are displaced during a mass layoff. The approach is motivated by the displaced worker literature, which has used mass layoffs as an exogenous source of job loss (Jacobson, LaLonde, and Sullivan, 1993). The results complement the main estimates in Section 4.3, as they rely on entirely distinct identification assumptions.

Following the literature, I identify a mass layoff event as a sharp and persistent contraction in firm size. Specifically, firm j experiences a mass layoff in quarter t if firm size, E_{jt} , declines by at least 30 percent in the following and subsequent year (i.e., $(E_{jk} - E_{jt})/E_{jt} \leq -0.3$ for $k \in \{t+4, t+8\}$). To address measurement issues, I require that the firm has between 50 and 5,000 employees in quarter t and I drop cases that are likely driven by changes in the firm identifier.³⁴ I define displaced workers as a subset of the intergenerational sample who have at least one year of tenure at firm j in quarter t , who separate during the mass layoff event, and who find a new job within two years. If an individual appears multiple times in the sample, I retain the first instance.

I estimate the following regression via ordinary least squares to compare the outcomes of young workers who are displaced in the same mass layoff event but subsequently take jobs either at their parent's employer or at a different firm:

$$\Delta y_{it+1} = \alpha + \beta D_{it+1} + \phi X_{it} + \lambda_{j(i,t)} + u_{it+1}, \quad (\text{D.1})$$

where individual i is employed at a firm that experiences a mass layoff in quarter t and subsequently finds a new job in $t+1$; D_{it+1} is an indicator for whether i joins their parent's employer in their first job after displacement (as before, I focus on the parent who is the primary earner); Δy_{it+1} is the difference between the log of quarterly earnings at the new job after displacement and the average log earnings in the four quarters prior to displacement; X_{it} is a vector of covariates defined prior to displacement, including the average of the outcome (e.g., log earnings) in the four quarters prior to displacement, parental earnings rank, fixed effects for age, and fixed effects for the interaction between race, sex, and year of birth; and $\lambda_{j(i,t)}$ is a fixed effect for the employer from which i was displaced in quarter t . Standard errors are clustered at the level of firm, $j(i, t)$. This specification takes the form of the estimator discussed by Dube et al. (2023).

The key identification assumption is that displaced workers who join their parent's employer are not systematically different from their observably similar displaced co-workers who do not. Two features of the research design lend credibility to this assumption. First, by comparing individuals employed at the same firm prior to displacement and controlling for pre-displacement earnings and demographic characteristics, the analysis accounts for a wide range of pre-displacement differences that could shape subsequent outcomes. Second, comparing workers displaced during the same mass layoff event mitigates concerns about endogenous job mobility and differential exposure to local economic conditions, as all workers are forced to search for new jobs in the same labor market.

Column 1 of Table A.4 presents estimates from a bivariate regression of y_{it} on D_{it+1} and shows that displaced workers who are subsequently re-employed at their parent's employer earn 12.9 log points less prior to displacement. These results are consistent with Table A.2 and show that

³⁴Firms are dropped from my sample if any worker who is there at time t experiences a change in the firm identifier (the "SEIN") but not a change in the job identifier (the "fid"). If a firm experiences multiple mass layoffs I only retain the first event.

individuals who find jobs through parental connections are negatively selected. Column 2 adds the firm fixed effect, $\lambda_{j(i,t)}$, and the difference falls to 3.7 log points, which shows that there is significantly less selection when comparing workers within the same firm.

Column 3 of Table A.4 presents estimates from a regression of Δy_{it+1} on D_{it+1} and $\lambda_{j(i,t)}$ and shows that displaced workers who subsequently find new jobs at their parent's employers experience a 22 log point gain in earnings relative to those who do not. Column 4 presents estimates from Equation D.1 and adds the full vector of covariates. The estimated earnings gains are quite stable at 21 log points, or 23 percent ($23 = (e^{0.208} - 1) \times 100$). Column 5 shows that working for a parent's employer leads individuals to work for firms that pay all workers 19 log points more, which is very similar to the estimated impacts on individual earnings.

Equation D.1 controls for level-differences in earnings prior to displacement, but it is possible that workers may be on differential trajectories. To investigate this, I estimate the following event-study specification:

$$y_{it} = \alpha_i + \sum_{k \neq -1} D_{it}^k \beta^k + \phi_{jt} + \gamma X_{it} + u_{it}, \quad (\text{D.2})$$

where i is the individual, t is the quarter relative to displacement, j is the employer prior to displacement, D_{it}^k is equal to one if the individual joined their parent's employer at their first job after being displaced and the displacement occurred k quarters ago as of quarter t , ϕ_{jt} is a fixed effect defined by the interaction between employer prior to displacement and the quarter relative to displacement, X_{it} is a quadratic in age, and u_{it} is a regression residual clustered at the level of j . The estimation sample is a balanced panel that includes the twelve quarters before and after the displacement event (inclusive of quarters with zero earnings).

Panel A of Figure A.9 presents the estimates from Equation D.2 and shows that individuals who join their parents' employers after being displaced earn significantly more than those who do not. The pre-displacement estimates show no sign of differential pre-trends leading up to the displacement, which supports the validity of the research design.³⁵ Panel B presents analogous estimates from a Poisson model.

Taken together, these results strengthen confidence in the main findings, as the instrumental variables strategy and the displaced workers analysis rely on distinct identification assumptions yet reach the same conclusion: working for a parent's employer significantly increases earnings, by providing access to higher-paying firms.

References

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- Dube, Arindrajit, Daniele Girardi, Oscar Jorda, and Alan M Taylor, "A local projections approach to difference-in-differences event studies," *National Bureau of Economic Research Working Paper No. 31184*, 2023.

³⁵The displaced workers separate in quarter 0 but could have earnings at the origin firm in this quarter. For this reason, the earnings losses are largest in quarter 1, which is the quarter after the job separation occurs.

Appendix E LATE vs. ATT

This section identifies two conditions under which the LATE is equivalent to the ATT. Let $Y_i(d, z)$ denote the potential outcome of individual i who has treatment status $D_i = d \in \{0, 1\}$ and instrument value $Z_i = z \in \{\underline{z}, \bar{z}\}$, where $\underline{z} < \bar{z}$. D_{zi} denotes the treatment status of i when $Z_i = z$. Furthermore, assume the following:

1. (Exclusion) $Y_i(d, \underline{z}) = Y_i(d, \bar{z}) \equiv Y_{di}$ for $d \in \{0, 1\}$,
2. (Independence) $\{Y_{0i}, Y_{1i}, D_{\underline{z}i}, D_{\bar{z}i}\} \perp\!\!\!\perp Z_i$,
3. (First Stage) $\mathbb{E}[D_{\bar{z}i} - D_{\underline{z}i}] > 0$, and
4. (Monotonicity) $D_{\underline{z}i} \leq D_{\bar{z}i} \forall i$.

Under these assumptions, the instrumental variables estimator identifies a LATE, which is the average treatment effect for the compliers: $\mathbb{E}[\beta_i | D_{\underline{z}i} < D_{\bar{z}i}]$, where $\beta_i = Y_{1i} - Y_{0i}$ denotes the individual-level treatment effect.

The ATT can be decomposed into a weighted average of the LATE and the average treatment effects for the always-takers (ATAT):

$$\begin{aligned}
 \underbrace{\mathbb{E}[\beta_i | D_i = 1]}_{\text{ATT}} &= \frac{\Pr(D_{\underline{z}i} < D_{\bar{z}i}, Z_i = \bar{z})}{\Pr(D_i = 1)} \mathbb{E}[\beta_i | D_{\underline{z}i} < D_{\bar{z}i}, Z_i = \bar{z}] \\
 &\quad + \frac{\Pr(D_{\underline{z}i} = 1)}{\Pr(D_i = 1)} \mathbb{E}[\beta_i | D_{\underline{z}i} = 1] \\
 &= \underbrace{\pi \mathbb{E}[\beta_i | D_{\underline{z}i} < D_{\bar{z}i}]}_{\text{LATE}} + \underbrace{(1 - \pi) \mathbb{E}[\beta_i | D_{\underline{z}i} = 1]}_{\text{ATAT}},
 \end{aligned} \tag{E.1}$$

where $\Pr(\cdot)$ denotes probability and $\pi = \frac{\Pr(D_{\underline{z}i} < D_{\bar{z}i}, Z_i = \bar{z})}{\Pr(D_i = 1)}$ denotes the share of treated individuals who are compliers. The second equality follows from independence, which implies that $\mathbb{E}[\beta_i | D_{\underline{z}i} < D_{\bar{z}i}, Z_i = \bar{z}] = \mathbb{E}[\beta_i | D_{\underline{z}i} < D_{\bar{z}i}]$, and monotonicity, which implies $\frac{\Pr(D_{\underline{z}i} < D_{\bar{z}i}, Z_i = \bar{z})}{\Pr(D_i = 1)} = 1 - \frac{\Pr(D_{\underline{z}i} = 1)}{\Pr(D_i = 1)}$. This decomposition highlights two cases under which the LATE is equal to the ATT.

Case 1. No Always-Takers. If there are no always-takers ($D_{\underline{z}i} = 0$ for all i) then all treated individuals are compliers ($\pi = 1$) and from Equation E.1 it follows that the LATE and ATT are equal.

Case 2. Equal Treatment Effects for Compliers and Always-Takers. If the average treatment effect among compliers and always-takers is equal ($\mathbb{E}[\beta_i | D_{\underline{z}i} < D_{\bar{z}i}] = \mathbb{E}[\beta_i | D_{\underline{z}i} = 1]$) then it immediately follows from Equation E.1 that the LATE and ATT are equal.

In the standard selection framework the second condition will not hold when there is selection on gains because treatment is determined by a single agent who weighs the benefits of treatment against its costs, which may be shifted by the instrument. In such a framework, the instrument and the treatment effects are linked through the agent's decision, and the LATE will depend on the values of the instrument. In my context, treatment status is determined by two distinct decisions: the parent's employer must make an offer, and the young worker must accept it. The multi-agent selection process potentially breaks the link between the instruments and the treatment effects.

To see why, consider a two-agent selection model in which the individual works for their parent's employer if and only if the employer makes them a job offer and they choose to accept the offer. The employer's decision to make an offer depends on the instruments and is defined as, $O_{zi} = \mathbb{1}\{\eta_i^O < z\}$. The child's decision to accept the offer depends on the treatment effects and is defined as, $A_i = \mathbb{1}\{\eta_i^A < \beta_i\}$. The variables η_i^O and η_i^A are offer and acceptance thresholds, respectively. I assume their values are not affected by treatment status or the instrument and I modify the independence assumption to also assume that $\{\eta_i^O, \eta_i^A\} \perp\!\!\!\perp Z_i$.³⁶ Treatment status is then defined as, $D_{zi} = O_{zi} \times A_i$.

The average treatment effect among compliers and always-takers will be equal if the firm's offer threshold is unrelated to the worker-side determinants of accepting an offer. Formally, if $\{\beta_i, \eta_i^A\} \perp\!\!\!\perp \eta_i^O$, then

$$\underbrace{\mathbb{E}[\beta_i \mid \eta_i^A < \beta_i, \underline{z} \leq \eta_i^O < \bar{z}]}_{\text{Compliers}} = \underbrace{\mathbb{E}[\beta_i \mid \eta_i^A < \beta_i, \eta_i^O < \underline{z}]}_{\text{Always-Takers}}. \quad (\text{E.2})$$

Because of the multi-agent nature of the selection problem, the LATE and ATT may be equivalent even in the presence of selection on gains. Appendix H develops a stylized behavioral model and discusses a specific case of Equation E.2.

³⁶Let $\eta_i^x(d, z)$ denote the potential outcome with treatment status $D_i = d$ and instrument value $Z_i = z$. I assume that $\eta_i^x = \eta_i^x(d, z)$ for $x \in \{O, A\}$.

Appendix F Estimating Marginal Treatment Effects

Following Cornelissen et al. (2016), let D_i denote treatment status and let p_i denote the propensity score. Treatment is determined by

$$D_i = \mathbb{1}\{p_i \geq U_{D_i}\},$$

where U_{D_i} denotes the individual's unobserved resistance to treatment. U_{D_i} is distributed uniformly on the unit interval.

I estimate marginal treatment effects by implementing the following steps.

First, I estimate the propensity score from the first stage described in Equation 4. The predicted value from this regression is the estimated propensity score and is denoted $\hat{p}_i$.

Second, I estimate the mean value of initial log earnings conditional on the estimated propensity score and the covariates listed in Equation 4. In the ideal implementation described in Section 4.3 of Cornelissen et al. (2016), the mean value would be modeled as a flexible function of $\hat{p}_i$ and would also include the linear interaction terms between $\hat{p}_i$ and all covariates. However, it is not computationally feasible to interact $\hat{p}_i$ with the fixed effects. Instead, I model the interaction terms with the firm and local labor market fixed effects using the AKM firm premium and the county-level unemployment rate. I estimate the following equation,

$$y_i = \pi + f(\hat{p}_i) + \phi x_{it-1} + \theta x_{it-1} \hat{p}_i + \omega \Psi_{p(i)} \hat{p}_i + \zeta u_{it} \hat{p}_i + \delta_{p(i)}^2 + \lambda_{l(p(i),t)}^2 + v_i, \quad (\text{F.1})$$

where $f(\cdot)$ is a third order polynomial, $\Psi_{p(i)}$ is the AKM firm pay premium associated with the parent's employer, and u_{it} is the commuting zone level unemployment rate. The main effects of $\Psi_{p(i)}$ and u_{it} are subsumed by the fixed effects. All other variables are defined in the description of Equation 4.

Third, I calculate the MTE by differentiating Equation F.1 with respect to $\hat{p}_i$ and evaluating the covariates at their sample means,

$$MTE(U_D) = \left. \frac{\partial f(p)}{\partial p} \right|_{p=U_D} + \theta \bar{x} + \omega \bar{\Psi} + \zeta \bar{u} \quad (\text{F.2})$$

Figure A.11 plots these MTE estimates against unobserved resistance to treatment. The support of the estimated propensity score determines the portion of the U_D distribution over which the MTE is directly identified. I therefore restrict the horizontal axis to the range of values between the 5th and 95th percentiles of the estimated propensity score.

References

Cornelissen, Thomas, Christian Dustmann, Anna Raute, and Uta Schönberg. "From LATE to MTE: Alternative methods for the evaluation of policy interventions." *Labour Economics* 41 (2016): 47-60.

Appendix G Approximation Methodology

To estimate $\text{cov}(D_i\beta_i, y_p)$ I rely on the following approximation:

$$\text{cov}(D_i\beta_i, y_p) \approx \mathbb{E}_r \left[\mathbb{E}[D_i|r_p] \mathbb{E}[\beta_i|r_p, D_i = 1] \mathbb{E}[y_p|r_p] \right] - \mathbb{E}[D_i] \mathbb{E}[\beta_i|D_i = 1] \mathbb{E}[y_p], \quad (\text{G.1})$$

where r_p is the quantile rank of parental earnings and $\mathbb{E}_r[\cdot]$ denotes an expectation over the quantile ranks of parental earnings.

To derive this expression, note that, $\text{cov}(D_i\beta_i, y_p) = \mathbb{E}[D_i\beta_i y_p] - \mathbb{E}[D_i\beta_i] \mathbb{E}[y_p]$. By iterated expectations,

$$\mathbb{E}[D_i\beta_i] = \mathbb{E}_D [\mathbb{E}[D_i\beta_i|D_i]] = \mathbb{E}[D_i] \mathbb{E}[\beta_i|D_i = 1] \quad (\text{G.2})$$

and

$$\mathbb{E}[D_i\beta_i y_p] = \mathbb{E}_r [\mathbb{E}[D_i\beta_i y_p|r_p]] \quad (\text{G.3})$$

Because the Pearson correlation coefficient is bounded between -1 and 1, it follows that,

$$\text{cov}(D_i\beta_i, y_p|r_p)^2 \leq \text{var}(D_i\beta_i|r_p) \times \text{var}(y_p|r_p) \quad (\text{G.4})$$

As the number of quantiles approaches the sample size, $\text{var}(y_p|r_p)$ approaches zero and the covariance term approaches zero. Thus,

$$\begin{aligned} \mathbb{E}[y_p D_i\beta_i|r_p] &= \mathbb{E}[y_p|r_p] \times \mathbb{E}[D_i\beta_i|r_p] + \text{cov}(D_i\beta_i, y_p|r_p) \\ &\approx \mathbb{E}[y_p|r_p] \times \mathbb{E}[D_i\beta_i|r_p] \end{aligned} \quad (\text{G.5})$$

where Equation G.4 suggests that $\text{cov}(D_i\beta_i, y_p|r_p)$ will be close to zero when conditioned on parental earnings ranks that are defined at a sufficiently high level of detail. Combining these pieces yields the approximation in Equation G.1.

I assess the performance of the approximation methodology by using the same methodology to approximate the observed IGE. By definition, $\rho(y_{ij}, y_p) = \frac{\text{cov}(y_{ij}, y_p)}{\text{var}(y_p)}$. The variance term, $\text{var}(y_p)$, is directly observed and I use the following approximation for the covariance term,

$$\text{cov}(y_{ij}, y_p) \approx \mathbb{E}_r \left[\mathbb{E}[y_p|r_p] \times \mathbb{E}[y_{ij}|r_p] \right] - \mathbb{E}[y_p] \times \mathbb{E}[y_{ij}] \quad (\text{G.6})$$

This approximation relies on the same assumption used to derive Equation G.1.

To validate the approximation, I show that the IGE based on the micro data, 0.136, is similar to estimates derived from the approximation, 0.140.

Appendix H Stylized Model

Consider a model in which the log earnings of individual i employed at firm $j(i)$ is:

$$y_i = h_i + f_{j(i)} + u_i, \quad (\text{H.1})$$

where h_i is the log of human capital, $f_{j(i)}$ is the firm pay premium, and u_i captures idiosyncratic determinants of earnings.

Using the potential outcomes framework, let $j^1(i)$ denote the employer of individual i if they work for their parent's employer, and let $j^0(i)$ denote their outside option, defined as the employer they would work for otherwise. The firm pay premium of the child's employer can be written as,

$$f_{j(i)} = f_{j^0(i)} + D_i \beta_i, \quad (\text{H.2})$$

where D_i is an indicator equal to one if the individual works for their parent's employer and $\beta_i = f_{j^1(i)} - f_{j^0(i)}$ is the effect of working for a parent's employer. The use of parental connections impacts earnings solely through the firm premium.

An individual's outside option depends on their human capital. Specifically, the labor market exhibits assortative matching between workers and firms, characterized by:

$$f_{j^0(i)} = \lambda h_i + \nu_i, \quad (\text{H.3})$$

where ν_i is an idiosyncratic error term. If $\lambda > 0$ then individuals with higher levels of human capital tend to match to employers that offer higher pay premiums. The same assortative matching process applies to parents.

The relationship between the human capital of the child and earnings of the parent is characterized by,

$$h_i = x + \theta y_p + \eta_i, \quad (\text{H.4})$$

where p denotes the parent of i , y_p denotes the parent's log earnings, η_i is an idiosyncratic error term, and x is a constant. Human capital is increasing in parental earnings if $\theta > 0$. Equation H.1 also characterizes the log earnings of the parents, so $y_p = h_p + f_{j(p)} + u_p$, where $f_{j(p)} = f_{j^1(i)}$. For simplicity, I assume that parents do not work for the employers of their parents (the grandparents of i): $D_p = 0$.³⁷

The child works for their parent's employer if and only if the firm decides to make a job offer and the child decides to accept that offer. Let O_i be equal to one if the parent's employer makes a job offer to the child and zero otherwise. The offer decision depends on hiring costs, c^O , labor demand shifter, $z_i \in \{\underline{z}, \bar{z}\}$ with $\underline{z} < \bar{z}$, and the human capital of the parent and the child: $O_i = \mathbb{1}\{c^O < \phi h_p + \gamma h_i + z_i\}$, where ϕ and γ could be positive or negative. I assume that z_i is a valid instrument that satisfies: $\{\eta_i, \nu_i, u_i, \nu_p, u_p, y_p\} \perp\!\!\!\perp z_i$. Let A_i be equal to one if the child would accept a job offer from the parent's firm. The child will choose to accept the offer if the earnings gains, β_i , exceed any costs, c^A : $A_i = \mathbb{1}\{c^A < \beta_i\}$. We can then write treatment status as a function of the instrument:

$$D_i(z) = \mathbb{1}\{c^O < \phi h_p + \gamma h_i + z\} \times \mathbb{1}\{c^A < \beta_i\}, \quad (\text{H.5})$$

³⁷This assumption simplifies the analysis and allows me to write the earnings benefits associated with working for the parent's employer as a function of parental earnings and unobserved error terms $\beta_i = (\frac{\lambda}{1+\lambda} - \lambda\theta)y_p + \frac{1}{1+\lambda}\nu_p - \frac{\lambda}{1+\lambda}u_p - \lambda x - \lambda\eta_i - \nu_i$.

where $D_i(z_i) = D_i$. Unlike the standard selection models, Equation H.5 illustrates that selection into treatment depends on the choices of multiple agents: the parent's employer and the child.

Combining equations H.1, H.2, H.3, and H.4 yields the following relationship between the log earnings of the child and the log earnings of their parents:

$$y_i = \alpha_1 + \alpha_2 y_p + D_i \beta_i + \epsilon_i, \quad (\text{H.6})$$

where $\epsilon_i = v_i + (1 + \lambda)\eta_i + u_i$, $\alpha_1 = (1 + \lambda)x$, and $\alpha_2 = (1 + \lambda)\theta$.

Regressing y_i on y_p yields an estimate of the intergenerational elasticity of earnings (IGE). My goal is to understand how the IGE would change if no one worked for the same employer as a parent; i.e., if $D_i = 0$ for all i . Because treatment effects may be heterogeneous and D_i may be correlated with ϵ_i , estimating Equation H.6 by OLS does not generally identify the parameter of interest.³⁸ For this reason, I rely on the approach discussed in Section 5.1.

As discussed in Section 5.1, estimating the counterfactual IGE requires an estimate of the ATT. An instrumental variables estimator that uses z_i as an instrument identifies a LATE, defined as $\mathbb{E}[\beta_i | D_i(\underline{z}) < D_i(\bar{z})]$. Appendix E identifies two cases in which the LATE is equal to the ATT. The stylized model provides some economic intuition underlying the second case, which is stated in the following proposition.

Proposition 1 *If $\phi h_p + \gamma h_i \equiv \eta_i^O$, with $\{z_i, \beta_i\} \perp \eta_i^O$, then:*

$$\underbrace{\mathbb{E}[\beta_i | D_i = 1]}_{\text{ATT}} = \underbrace{\mathbb{E}[\beta_i | D_i(\underline{z}) < D_i(\bar{z})]}_{\text{LATE}} \quad (\text{H.7})$$

Proof 1 $\{z_i, \beta_i\} \perp \eta_i^O$ implies that $\{\beta_i, A_i\} \perp O_i$, which implies:

$$\mathbb{E}[\beta_i | A_i = 1] = \mathbb{E}[\beta_i | A_i = 1, O_i(\underline{z}) < O_i(\bar{z})] = \mathbb{E}[\beta_i | A_i = 1, O_i(\underline{z}) = 1]. \quad (\text{H.8})$$

It follows that:

$$\begin{aligned} \mathbb{E}[\beta_i | D_i = 1] &= \frac{\Pr(D_i(\underline{z}) < D_i(\bar{z}), z_i = \bar{z})}{\Pr(D_i = 1)} \mathbb{E}[\beta_i | D_i(\underline{z}) < D_i(\bar{z}), z_i = \bar{z}] \\ &\quad + \frac{\Pr(D_i(\underline{z}) = 1)}{\Pr(D_i = 1)} \mathbb{E}[\beta_i | D_i(\underline{z}) = 1] \\ &= \mathbb{E}[\beta_i | A_i = 1, O_i(\underline{z}) < O_i(\bar{z})] \\ &= \mathbb{E}[\beta_i | D_i(\underline{z}) < D_i(\bar{z})], \end{aligned} \quad (\text{H.9})$$

where $\Pr(\cdot)$ denotes probability. The first equality follows from Equation H.5, which implies that $\Pr(D_i = 1) = \Pr(D_i(\underline{z}) < D_i(\bar{z}), z_i = \bar{z}) + \Pr(D_i(\underline{z}) = 1)$. The second equality follows from Equation H.8.

Proposition 1 shows that, if the firm's offer decision is unrelated to the worker-side determinants of accepting an offer, then the ATT and LATE will be equal. One case in which $\{z_i, \beta_i\} \perp \eta_i^O$ will

³⁸To see the relationship between D_i and ϵ_i note that $\epsilon_i = v_i + (1 + \lambda)\eta_i + u_i$, $O_i = \mathbb{1}\{c^O < (\frac{\phi}{1+\lambda} + \gamma\theta)y_p - \frac{\phi}{1+\lambda}(v_p + u_p) + \gamma(x + \eta_i) + z_i\}$, and $A_i = \mathbb{1}\{c^A + \lambda x + \lambda\eta_i + v_i < (\frac{\lambda}{1+\lambda} - \lambda\theta)y_p + (\frac{\lambda}{1+\lambda})(v_p/\lambda - u_p)\}$.

be true is if the offer decision is unrelated to the human capital of the parent ($\phi = 0$) and the human capital of the child ($\gamma = 0$) implying that $\eta_i^O = 0$. Such a case is plausible given that the sign of both parameters is theoretically ambiguous. ϕ might be positive if higher-ability parents have more control over the hiring process because they hold leadership positions, or negative if lower-ability parents work at firms that rely more heavily on networks in the hiring process. γ may be positive if firms are more likely to make a job offer to high-ability workers, or negative if parents exert more effort to procure job opportunities for low-ability children.

The following proposition shows that the intergenerational transmission of employers has a theoretically ambiguous effect on intergenerational mobility.

Proposition 2 *Consider a deterministic case of the model by letting z_i , η_i , v_i and u_i be equal to zero and let $c^A \geq 0$ and $\lambda \geq 0$. Then the following statements are true:*

- if $\frac{1}{1+\lambda} > \theta$ and $\phi > -\theta\gamma(1+\lambda)$ then $\rho(y_i, y_p) \geq \rho(y_i^0, y_p)$
- if $\frac{1}{1+\lambda} < \theta$ and $\phi < -\theta\gamma(1+\lambda)$ then $\rho(y_i, y_p) \leq \rho(y_i^0, y_p)$,

where $y_i^d = h_i + f_{j(i)}^d + u_i$ denotes the potential outcome under treatment status $d \in \{0, 1\}$ and $\rho(y_i, y_p)$ denotes the coefficient from a regression of y_i on y_p .

Proof 2 *To prove the results it is useful to start by noting the implications of the deterministic setting (η_i , v_i , u_i and z_i are set to zero) for the following expressions,*

$$\begin{aligned} O_i &= \mathbb{1}\{c^O < (\frac{\phi}{1+\lambda} + \theta\gamma)y_p + \gamma x\} \\ A_i &= \mathbb{1}\{c^A < (\frac{\lambda}{1+\lambda} - \lambda\theta)y_p - \lambda x\} \\ \beta_i &= (\frac{\lambda}{1+\lambda} - \lambda\theta)y_p - \lambda x \end{aligned} \tag{H.10}$$

It is straightforward to show that $\text{cov}(\beta_i, y_p) = (\frac{\lambda}{1+\lambda} - \lambda\theta) \text{var}(y_p)$. In the first case, when $\frac{1}{1+\lambda} > \theta$ and $\phi > -\theta\gamma(1+\lambda)$, it immediately follows that $\frac{\partial \beta_i}{\partial y_p} > 0$, $\frac{\partial O_i}{\partial y_p} \geq 0$, $\frac{\partial A_i}{\partial y_p} \geq 0$ and $\frac{\partial D_i}{\partial y_p} \geq 0$. Since β_i and D_i are both weakly increasing in y_p , and $c^A \geq 0$ implies $D_i\beta_i \geq 0$, the product $D_i\beta_i$ is also weakly increasing in y_p . Thus, $\text{cov}(\beta_i, y_p)$ and $\text{cov}(D_i\beta_i, y_p)$ have the same sign, which implies that, $\text{cov}(D_i\beta_i, y_p) \geq 0$. The proof for the second case uses the same logic.

Proposition 2 highlights two competing forces. On the one hand, high-income parents are best able to procure high-paying job offers for their children. On the other hand, children from low-income households have lower levels of human capital and are more reliant on their parents to find a higher-paying job. Thus, while the empirical evidence suggests that the intergenerational transmission of employers increases the intergenerational persistence in earnings, this conclusion might differ in other contexts depending on the characteristics of the labor market and the human capital accumulation process.

H.1 Extension with Parental Investment in Human Capital

In the spirit of the canonical models of intergenerational mobility from Becker and Tomes (1979, 1986), I extend the baseline model to allow parents to make decisions regarding the optimal investments of the human capital of their children.

For tractability I focus on the deterministic setting (z_i , η_i , v_i , and u_i are equal to zero) and assume that children only accept job offers from their parent's employer when the earnings benefits are positive ($c^A \geq 0$) and that there is assortative matching between workers and firms ($\lambda > 0$). I maintain the assumptions underlying equations H.1, H.2, and H.3. I assume that $\gamma \leq 0$, which simplifies the analysis by making D_i and β_i decreasing in the human capital of the child. I do not impose the assumption stated in Equation H.4, because the goal of this section is to derive the relationship between parental earnings and the human capital of the child as the result of optimizing behavior on the part of the parents. For notation, I use lowercase letters to denote the log of uppercase variables (for example, $h_i = \log(H_i)$).

Parents care about their current period consumption, C_p , and the total financial resources of their children, which depends on the earnings of the children, Y_i , and bequests, B_i , plus interest accrued at rate R . Parents solve the following problem:

$$\max_{C_p, S_i, B_i} \{v(C_p) + u(Y_i + RB_i)\} \text{ subject to } C_p + S_i + B_i \leq Y_p \quad (\text{H.11})$$

where S_i represents investment in the human capital of the children and $u(\cdot)$ and $v(\cdot)$ are continuous functions that both have the following properties: $u'(\cdot) > 0$, $u''(\cdot) < 0$ and $u'(0) = \infty$. This setup assumes that there are no credit constraints.

While there are a number of ways to generate intergenerational persistence in earnings in the absence of credit constraints, I follow Becker et al. (2018) and assume that there are complementarities between the human capital of the parent and the production of human capital of the child. Specifically, investment translates into human capital according to the following production function, $H_i = H_p^\sigma S_i^\alpha$. Intuitively, this captures the fact that investments in human capital might be more productive if made by parents with higher ability. I also assume that $0 < \alpha(1 + \lambda) < 1$ which implies that there are diminishing returns to parental investment.

Assuming an interior solution for bequests, the optimal level of investment in human capital is defined by the level at which the marginal rate of return is equal to the interest rate, $\frac{\partial Y_i}{\partial S_i} = R$. The expression determining optimal investment is:

$$\alpha(1 + \lambda)H_p^{\sigma(1+\lambda)}S_i^{\alpha(1+\lambda)-1}\exp\{D_i\beta_i\} + H_p^{\sigma(1+\lambda)}S_i^{\alpha(1+\lambda)}\exp\{D_i\beta_i\}\frac{\partial D_i\beta_i}{\partial S_i} = R, \quad (\text{H.12})$$

where the left-hand side represents the marginal returns to investments in human capital and the right-hand side represents the marginal returns to bequests.³⁹

Start by considering the case where parents ignore the role of connections when making

³⁹Technically, the derivative $\frac{\partial \exp\{D_i\beta_i\}}{\partial S_i}$ may not exist because $D_i\beta_i$ may be discontinuous in S_i . I ignore this possibility to simplify the analysis. However, it would be straightforward to address this issue by considering a nondeterministic setting in which parents optimize over the expectation of $D_i\beta_i$.

investment decisions. The solution to Equation H.12 under the assumption that $D_i = 0$ is:

$$s_i^{naive} = \frac{\log(\alpha(1+\lambda)) + \sigma(1+\lambda)h_p - r}{1 - \alpha(1+\lambda)}. \quad (\text{H.13})$$

Next, consider the case where parents fully account for the impact of connections. Then Equation H.12 implies the optimal investment is defined by the following implicit condition:

$$s_i^* = s_i^{naive} + \frac{D_i\beta_i}{1 - \alpha(1+\lambda)} + \frac{\log(\alpha(1+\lambda) - \lambda\alpha D_i + S_i^*\beta_i \frac{\partial D_i}{\partial S_i}) - \log(\alpha(1+\lambda))}{1 - \alpha(1+\lambda)}, \quad (\text{H.14})$$

where D_i and β_i are functions of S_i^* . The log optimal level of investment is the naive solution plus two additional terms.

The first term in Equation H.14 captures the extent to which parental connections raise the marginal returns to human capital investment by providing access to higher-paying firms. When children use parental connections to obtain jobs at firms with higher wage premia, each additional unit of human capital leads to a larger increase in earnings.⁴⁰ This channel increases optimal investment because $D_i\beta_i \geq 0$ and $\alpha(1+\lambda) < 1$, implying that

$$\frac{D_i\beta_i}{1 - \alpha(1+\lambda)} \geq 0. \quad (\text{H.15})$$

The second term in Equation H.14 reflects a countervailing mechanism: parental connections reduce the return to human capital investment by weakening the penalty associated with low human capital. Human capital increases earnings in part by helping workers obtain jobs at higher-paying firms. However, when parents expect their children to access such firms through family connections, the marginal value of investing in human capital falls. This force reduces optimal investment. In particular, $\lambda\alpha D_i \geq 0$ and $S_i^*\beta_i \frac{\partial D_i}{\partial S_i} \leq 0$, implying that:⁴¹

$$\frac{\log(\alpha(1+\lambda) - \lambda\alpha D_i + S_i^*\beta_i \frac{\partial D_i}{\partial S_i}) - \log(\alpha(1+\lambda))}{1 - \alpha(1+\lambda)} \leq 0. \quad (\text{H.16})$$

Because these two forces operate in opposite directions, the effect of parental connections on human capital investment is theoretically ambiguous. On the one hand, parental connections can increase the marginal return to investment by giving children access to firms where human capital is rewarded more highly. On the other hand, those same connections can reduce investment incentives by insulating children from the earnings penalty associated with low human capital. Hence, policies that restrict firms from hiring the children of current employees could lead parents to invest either more or less in their children's human capital, depending on which force dominates.

The implications for intergenerational mobility are similarly ambiguous. To see why, consider the case in which children from higher-income families are more likely to work for a parent's employer

⁴⁰Different assumptions could lead to alternative conclusions. For example, both Corak and Piraino (2010) and Magruder (2010) assume that the effect of networks on earnings is additive in levels, which leads them to conclude that parental investment decisions are unaffected by the presence of parental labor market networks.

⁴¹ $\frac{\partial D_i}{\partial S_i} \leq 0$ is a consequence of the assumption that $\gamma \leq 0$. Relaxing this assumption would not affect the conclusion that the implications for investment are theoretically ambiguous.

and experience larger earnings gains when they do. The first force described above amplifies the earnings disparities between children from high- and low-income families: by increasing the marginal return to human capital investment, parental connections induce greater investment among high-income families whose children have access to higher-paying jobs through those connections. The second force works in the opposite direction. Because children from high-income families can obtain high-paying jobs through their parents' connections, their parents have less incentive to invest in their human capital. As a result, decisions to invest in human capital could either amplify or attenuate the direct effects of parental connections on intergenerational mobility.

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